Volume Ii Reference

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Formalizing the Number Systems

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Capstone

Identity, Equality, and Equivalence

0.1 Identity, Equality, and Equivalence

0.2 Equality

Definition 0.2.1 (Identity)**.** Two expressions x and y are *identical* when they denote the same object.

Definition 0.2.2 (Equality on a Domain)**.** Let A be a domain. *Equality on A* is the relation, written $=$, that holds exactly when two terms or constructions denote the same element of A.

Definition 0.2.3 (Definitional and Propositional Equality)**.** *Definitional equality* records an introduced meaning, written with $:=$. *Propositional equality* is a statement of sameness inside the theory, written with $=$.

0.3 Properties of Equality

Theorem 0.3.1 (Symmetry of Equality)**.**

$$\forall x\,\forall y,\ (x = y \implies y = x).$$

Theorem 0.3.2 (Transitivity of Equality)**.**

$$\forall x\,\forall y\,\forall z,\ \big(x = y \land y = z \implies x = z\big).$$

Corollary 0.3.3 (Equality Is an Equivalence Relation)**.** *The equality relation is reflexive, symmetric, and transitive.*

0.4 Substitution

Proposition 0.4.1 (Substitution Preserves Predicates)**.**

$$x = y \implies \big(P(x) \iff P(y)\big).$$

Proposition 0.4.2 (Substitution Preserves Relations on the Left)**.**

$$x = y \implies \big(R(x, z) \iff R(y, z)\big).$$

Proposition 0.4.3 (Substitution Preserves Relations on the Right)**.**

$$x = y \implies \big(R(z, x) \iff R(z, y)\big).$$

Proposition 0.4.4 (Substitution Preserves Relations)**.**

$$x = x' \land y = y' \implies \big(R(x, y) \iff R(x', y')\big).$$

Proposition 0.4.5 (Substitution Preserves Functions)**.**

$$x = y \Longrightarrow f(x) = f(y).$$

Proposition 0.4.6 (Left Congruence for Operations)**.**

$$x = x' \Longrightarrow x \, O \, y = x' \, O \, y.$$

Proposition 0.4.7 (Right Congruence for Operations)**.**

$$y = y' \Longrightarrow x \, O \, y = x \, O \, y'.$$

Proposition 0.4.8 (Full Congruence for Operations)**.**

$$x = x' \wedge y = y' \Longrightarrow x \, O \, y = x' \, O \, y'.$$

Proposition 0.4.9 (Congruence with Respect to Equality Is Automatic)**.** *Every function, predicate, relation, and operation respects equality in its displayed arguments.*

0.5 Equivalence

Definition 0.5.1 (Equivalence Relation)**.** Let A be a set. A relation $\sim$ on A is an *equivalence relation* if it is reflexive, symmetric, and transitive.

Corollary 0.5.2 (Equality Is the Finest Equivalence)**.** *Let $\sim$ be an equivalence relation on A. For all $a, b \in A$,*

$$a = b \Longrightarrow a \sim b.$$

Definition 0.5.3 (Equivalence Class)**.** Let $\sim$ be an equivalence relation on A. The *equivalence class* of $a \in A$ is

$$[a] := \{x \in A \mid x \sim a\}.$$

Theorem 0.5.4 (Equivalence Classes Partition the Domain)**.** *The equivalence classes of an equivalence relation on A are nonempty, cover A, and are pairwise disjoint.*

Proposition 0.5.5 (Representatives Related iff Classes Equal)**.**

$$a \sim b \Longleftrightarrow [a] = [b].$$

Canonical Projection

Definition 0.5.6 (Canonical Projection)**.** Let $\sim$ be an equivalence relation on A. The *canonical projection* is the map

$$\pi : A \to A/{\sim}, \qquad \pi(a) := [a].$$

Proposition 0.5.7 (Projection Is Surjective)**.** *The canonical projection $\pi : A \to A/{\sim}$ is surjective.*

Proposition 0.5.8 (Projection Identifies Exactly the Equivalent Elements)**.** *For all $a, b \in A$,*

$$\pi(a) = \pi(b) \Longleftrightarrow a \sim b.$$

Theorem 0.5.9 (Existence of a Quotient)**.** *For any equivalence relation $\sim$ on A, the quotient set $A/{\sim}$ exists as the set of equivalence classes, and the canonical projection*

$$\pi : A \to A/{\sim}$$

is well-defined.

Induced Structure on a Quotient

Definition 0.5.10 (Operation Respects an Equivalence)**.** Let $O : A \times A \to A$ be a binary operation and let $\sim$ be an equivalence relation on A. The operation O *respects* $\sim$ if

$$\forall a, a', b, b' \in A,\ \big(a \sim a' \wedge b \sim b'\big) \Longrightarrow \big(a\, O\, b\big) \sim \big(a'\, O\, b'\big).$$

Definition 0.5.11 (Left and Right Respect)**.** The operation O *respects* $\sim$ on the left if

$$a \sim a' \Longrightarrow (a\, O\, b) \sim (a'\, O\, b),$$

and respects $\sim$ on the right if

$$b \sim b' \Longrightarrow (a\, O\, b) \sim (a\, O\, b').$$

Lemma 0.5.12 (Respect Splits)**.** *A binary operation O respects $\sim$ if and only if it respects $\sim$ on the left and on the right.*

Theorem 0.5.13 (Induced Operation on a Quotient)**.** *If $O : A \times A \to A$ respects $\sim$, then*

$$[a]\, \overline{O}\, [b] := [a\, O\, b]$$

defines a well-defined binary operation on $A/{\sim}$.

Theorem 0.5.14 (Induced Unary Operation)**.** *Let $f : A \to A$. If*

$$a \sim a' \Longrightarrow f(a) \sim f(a'),$$

then

$$\overline{f}([a]) := [f(a)]$$

defines a well-defined unary operation on $A/{\sim}$.

Definition 0.5.15 (Predicate Respects an Equivalence)**.** Let P be a predicate on A. The predicate P *respects* $\sim$ if

$$\forall a, a' \in A,\ a \sim a' \Longrightarrow \big(P(a) \Longleftrightarrow P(a')\big).$$

Theorem 0.5.16 (Induced Predicate on a Quotient)**.** *If P respects $\sim$, then*

$$\overline{P}([a]) :\Longleftrightarrow P(a)$$

defines a well-defined predicate on $A/{\sim}$.

Definition 0.5.17 (Relation Respects an Equivalence)**.** Let R be a binary relation on A. The relation R *respects* $\sim$ if

$$\forall a, a', b, b' \in A,\ \big(a \sim a' \wedge b \sim b'\big) \Longrightarrow \big(R(a, b) \Longleftrightarrow R(a', b')\big).$$

Theorem 0.5.18 (Induced Relation on a Quotient)**.** *If R respects $\sim$, then*

$$\overline{R}([a], [b]) :\Longleftrightarrow R(a, b)$$

defines a well-defined binary relation on $A/{\sim}$.

Theorem 0.5.19 (Induced Map out of a Quotient)**.** *Let $g : A \to B$. If*

$$a \sim a' \Longrightarrow g(a) = g(a'),$$

then there is a unique well-defined map

$$\overline{g} : A/{\sim} \to B$$

such that

$$\overline{g}([a]) = g(a) \qquad \textit{and} \qquad \overline{g} \circ \pi = g,$$

where $\pi : A \to A/{\sim}$ is the canonical projection.

Theorem 0.5.20 (Induced Partial Operation on a Quotient). *Let $D \subseteq A$ be saturated under $\sim$, meaning*

$$a \in D \wedge a \sim a' \Longrightarrow a' \in D.$$

If $f : D \to A$ respects $\sim$ on D, then

$$\overline{f}([a]) := [f(a)]$$

is a well-defined partial operation on $A/\sim$ with domain

$$D/\sim := \{[a] \in A/\sim : a \in D\}.$$

Proposition 0.5.21 (Representative Independence Is Necessary). *If a rule on classes*

$$[a]\, \overline{O}\, [b] := [a\, O\, b]$$

is well-defined on $A/\sim$, then O respects $\sim$.

Lemma 0.5.22 (Distinct Equivalence Classes Are Disjoint).

$$[a] \neq [b] \Longrightarrow [a] \cap [b] = \emptyset.$$

Proofs

Capstone

Arithmetic Operations and Relations

0.6 Arithmetic Operations and Relations

0.7 Closure

Definition 0.7.1 (Unary Arithmetic Operation)**.** Let K be a domain. A *unary arithmetic operation* on K is a function

$$O : K \to K.$$

Definition 0.7.2 (Binary Arithmetic Operation)**.** Let K be a domain. A *binary arithmetic operation* on K is a function

$$O : K \times K \to K.$$

Definition 0.7.3 (Unary Closure)**.** A unary operation O is *closed on K* if applying it to an element of K produces an element of K.

Definition 0.7.4 (Binary Closure)**.** A binary operation O is *closed on K* if combining two elements of K produces an element of K.

Definition 0.7.5 (Operation Well-Definedness)**.** Let $\sim$ be an equivalence relation on A. A binary operation O respects $\sim$ if

$$a \sim a' \wedge b \sim b' \Longrightarrow a\,O\,b \sim a'\,O\,b'.$$

Lemma 0.7.6 (Left Respect)**.** *If a binary operation respects $\sim$, then*

$$a \sim a' \Longrightarrow a\,O\,b \sim a'\,O\,b.$$

Lemma 0.7.7 (Right Respect)**.** *If a binary operation respects $\sim$, then*

$$b \sim b' \Longrightarrow a\,O\,b \sim a\,O\,b'.$$

Theorem 0.7.8 (Induced Operation)**.** *The operation*

$$[a]\,\overline{O}\,[b] := [a\,O\,b]$$

on $A/{\sim}$ is well-defined if and only if O respects $\sim$.

0.8 Identity Elements

Definition 0.8.1 (Left Identity)**.**

$$\forall x,\ e_L\,O\,x = x.$$

Definition 0.8.2 (Right Identity)**.**

$$\forall x,\ x\,O\,e_R = x.$$

Definition 0.8.3 (Two-Sided Identity)**.** An element e is a *two-sided identity* for O if it is both a left identity and a right identity.

Definition 0.8.4 (Absorbing Element). An element z is an *absorbing element* for a binary operation O if

$$z \, O \, x = z \quad \text{and} \quad x \, O \, z = z$$

for every element x in the domain of the operation.

Lemma 0.8.5 (Left and Right Identities Coincide). *If e_L is a left identity and e_R is a right identity for the same operation, then $e_L = e_R$.*

Theorem 0.8.6 (Uniqueness of Identity). *A two-sided identity for a binary operation is unique.*

0.9 Inverse Operations

Definition 0.9.1 (Left Inverse). An element x' is a left inverse of x with respect to identity e if

$$x' \, O \, x = e.$$

Definition 0.9.2 (Right Inverse). An element x'' is a right inverse of x with respect to identity e if

$$x \, O \, x'' = e.$$

Definition 0.9.3 (Two-Sided Inverse). An element y is a two-sided inverse of x if

$$y \, O \, x = e \quad \text{and} \quad x \, O \, y = e.$$

Lemma 0.9.4 (Left and Right Inverses Coincide). *In an associative operation with identity, a left inverse and a right inverse of the same element are equal.*

Theorem 0.9.5 (Uniqueness of Inverse). *In an associative operation with identity, inverses are unique.*

Definition 0.9.6 (Right Solvability). Right solvability means

$$\forall x, y \, \exists z, \; x = y \, O \, z,$$

Definition 0.9.7 (Left Solvability). Left solvability means

$$\forall x, y \, \exists z, \; x = z \, O \, y.$$

Definition 0.9.8 (Two-Sided Solvability). Two-sided solvability means both left solvability and right solvability hold.

Proposition 0.9.9 (Solvability and Inverses). *For an associative operation with a two-sided identity, solvability is equivalent to the existence of inverses.*

0.10 Algebraic Laws

Definition 0.10.1 (Associativity).

$$\forall x, y, z, \; x \, O \, (y \, O \, z) = (x \, O \, y) \, O \, z.$$

Theorem 0.10.2 (General Associativity). *If O is associative, then every bracketing of $x_1 \, O \cdots O \, x_n$ has the same value.*

Definition 0.10.3 (Commutativity).

$$\forall x, y, \; x \, O \, y = y \, O \, x.$$

Theorem 0.10.4 (General Commutativity). *If O is associative and commutative, then any finite reordering of $x_1 \, O \cdots O \, x_n$ has the same value.*

Definition 0.10.5 (Left Distributivity)**.** The operation $\otimes$ is left distributive over $\oplus$ if

$$x \otimes (y \oplus z) = (x \otimes y) \oplus (x \otimes z),$$

Definition 0.10.6 (Right Distributivity)**.** The operation $\otimes$ is right distributive over $\oplus$ if

$$(y \oplus z) \otimes x = (y \otimes x) \oplus (z \otimes x),$$

Definition 0.10.7 (Two-Sided Distributivity)**.** The operation $\otimes$ is two-sided distributive over $\oplus$ if it is both left distributive and right distributive.

Definition 0.10.8 (Left Cancellation)**.** An operation has left cancellation if

$$x \, O \, y = x \, O \, z \Longrightarrow y = z,$$

Definition 0.10.9 (Right Cancellation)**.** An operation has right cancellation if

$$y \, O \, x = z \, O \, x \Longrightarrow y = z.$$

Definition 0.10.10 (Two-Sided Cancellation)**.** An operation has two-sided cancellation if it has both left cancellation and right cancellation.

Proposition 0.10.11 (Invertibility Implies Left Cancellation)**.** *In an associative operation with identity, invertible elements satisfy left cancellation.*

Proposition 0.10.12 (Invertibility Implies Cancellation)**.** *In an associative operation with identity, invertible elements satisfy two-sided cancellation.*

Proposition 0.10.13 (Invertibility Implies Right Cancellation)**.** *In an associative operation with identity, invertible elements satisfy right cancellation.*

0.11 Derived Laws

Proposition 0.11.1 (Left Zero Absorption)**.** *When multiplication distributes over addition and* 0 *is the additive identity,*

$$0 \cdot a = 0.$$

Proposition 0.11.2 (Right Zero Absorption)**.** *When multiplication distributes over addition and* 0 *is the additive identity,*

$$a \cdot 0 = 0.$$

Proposition 0.11.3 (Zero Absorption)**.** *When multiplication distributes over addition and* 0 *is the additive identity,*

$$0 \cdot a = 0 \qquad \textit{and} \qquad a \cdot 0 = 0.$$

Proposition 0.11.4 (Left Sign Rule)**.** *When multiplication distributes over addition and additive inverses exist,*

$$(-a) \cdot b = -(a \cdot b).$$

Proposition 0.11.5 (Right Sign Rule)**.** *When multiplication distributes over addition and additive inverses exist,*

$$a \cdot (-b) = -(a \cdot b).$$

Proposition 0.11.6 (Sign Rule)**.** *When multiplication distributes over addition and additive inverses exist,*

$$(-a) \cdot b = -(a \cdot b) \qquad \textit{and} \qquad a \cdot (-b) = -(a \cdot b).$$

Corollary 0.11.7 (Negative One Rule)**.** *When multiplication distributes over addition and additive inverses exist,*

$$(-1) \cdot a = -a.$$

Proposition 0.11.8 (Product of Negatives)**.** *When multiplication distributes over addition and additive inverses exist,*

$$(-a) \cdot (-b) = a \cdot b.$$

Definition 0.11.9 (Zero Divisor)**.** Let K be an arithmetic system with multiplication and zero. A nonzero element $a \in K$ is a *zero divisor* if there exists a nonzero element $b \in K$ such that $ab = 0$.

Theorem 0.11.10 (No Zero Divisors)**.** *In an arithmetic system with no nonzero zero divisors,*

$$a \cdot b = 0 \implies a = 0 \vee b = 0.$$

0.12 Order Relations

Definition 0.12.1 (Reflexive Relation)**.**

$$\forall x,\ xRx.$$

Definition 0.12.2 (Irreflexive Relation)**.**

$$\forall x,\ \neg(xRx).$$

Definition 0.12.3 (Symmetric Relation)**.**

$$\forall x, y,\ xRy \implies yRx.$$

Definition 0.12.4 (Antisymmetric Relation)**.**

$$\forall x, y,\ (xRy \wedge yRx) \implies x = y.$$

Definition 0.12.5 (Asymmetric Relation)**.**

$$\forall x, y,\ xRy \implies \neg(yRx).$$

Definition 0.12.6 (Transitive Relation)**.**

$$\forall x, y, z,\ (xRy \wedge yRz) \implies xRz.$$

Definition 0.12.7 (Total Relation)**.**

$$\forall x, y,\ xRy \vee yRx.$$

Definition 0.12.8 (Trichotomy)**.** The relation $<$ satisfies *trichotomy* if exactly one of

$$x < y, \qquad x = y, \qquad y < x$$

holds for every pair x, y.

0.13 Order Compatibility

Definition 0.13.1 (Right Monotony)**.** Right monotony means

$$yRz \implies (x\,O\,y)R(x\,O\,z),$$

Definition 0.13.2 (Left Monotony)**.** Left monotony means

$$yRz \implies (y\,O\,x)R(z\,O\,x).$$

Definition 0.13.3 (Two-Sided Monotony)**.** Two-sided monotony means both left monotony and right monotony hold.

Definition 0.13.4 (Right Order Reflection)**.** Right reflection means

$$(x \, O \, y) R (x \, O \, z) \Longrightarrow yRz,$$

Definition 0.13.5 (Left Order Reflection)**.** Left reflection means

$$(y \, O \, x) R (z \, O \, x) \Longrightarrow yRz.$$

Definition 0.13.6 (Two-Sided Order Reflection)**.** Two-sided order reflection means both left order reflection and right order reflection hold.

Definition 0.13.7 (Positive cone)**.** Let K be an arithmetic system with addition, multiplication, and zero. A subset $P \subset K$ is called a *positive cone* if it satisfies:

1. if $a, b \in P$, then $a + b \in P$;
2. if $a, b \in P$, then $ab \in P$;
3. for every $a \in K$, exactly one of the following holds:

$$a \in P, \qquad a = 0, \qquad -a \in P.$$

The strict order associated to P is

$$a < b \iff b - a \in P.$$

Definition 0.13.8 (Monotone Map)**.**

$$x \leq y \Longrightarrow f(x) \leq f(y).$$

Definition 0.13.9 (Strictly Monotone Map)**.**

$$x < y \Longrightarrow f(x) < f(y).$$

Proposition 0.13.10 (Addition Preserves Order on the Left)**.**

$$y < z \Longrightarrow x + y < x + z$$

in ordered arithmetic systems where addition is strictly order-preserving.

Proposition 0.13.11 (Addition Preserves Order on the Right)**.**

$$y < z \Longrightarrow y + x < z + x$$

in ordered arithmetic systems where addition is strictly order-preserving.

Proposition 0.13.12 (Addition Preserves Order)**.** *In ordered arithmetic systems where addition is strictly order-preserving, addition preserves order on both sides.*

Proposition 0.13.13 (Multiplication Preserves Order on the Left)**.**

$$0 < x \wedge y < z \Longrightarrow x \cdot y < x \cdot z$$

in ordered arithmetic systems where multiplication by positive elements is strictly order-preserving.

Proposition 0.13.14 (Multiplication Preserves Order on the Right)**.**

$$0 < x \wedge y < z \Longrightarrow y \cdot x < z \cdot x$$

in ordered arithmetic systems where multiplication by positive elements is strictly order-preserving.

Proposition 0.13.15 (Multiplication Preserves Order)**.** *In ordered arithmetic systems where multiplication by positive elements is strictly order-preserving, multiplication preserves order on both sides.*

0.14 Composed Relations

Definition 0.14.1 (Strict Partial Order). A *strict partial order* is an irreflexive and transitive relation.

Definition 0.14.2 (Strict Total Order). A *strict total order* is a strict partial order satisfying trichotomy.

0.15 Structure-Preserving Maps

Definition 0.15.1 (Homomorphism). A map f is a *homomorphism* for an operation O when

$$f(x\,O\,y) = f(x)\,O'\,f(y).$$

For ring-like structures it preserves both addition and multiplication, and preserves 1 when the language includes a distinguished multiplicative identity.

Definition 0.15.2 (Injective Homomorphism). An *injective homomorphism* is a homomorphism whose underlying function is injective.

Definition 0.15.3 (Order Embedding).

$$xRy \iff f(x)R'f(y).$$

Definition 0.15.4 (Structure Embedding). A *structure embedding* is an injective homomorphism that also embeds the relevant order relation.

Proposition 0.15.5 (Composition of Homomorphisms). *The composition of two homomorphisms is a homomorphism.*

Corollary 0.15.6 (Identity Map Is a Homomorphism). *The identity map on a structure is a homomorphism from that structure to itself.*

0.16 Algebraic Structures

0.17 Field and Ordered-Field Laws

Proofs

Capstone

Constructing a Number System

0.18 Ground Rules

Construction Obligations

Definition 0.18.1 (Number-System Construction Target)**.** A *number-system construction target* is the specified algebraic and order structure that a proposed system is meant to satisfy. The target fixes the required carrier, operations, relations, distinguished elements, and laws.

Definition 0.18.2 (Construction Data)**.** The *construction data* for a number system consists of the raw carrier objects, the intended equality or equivalence relation, the distinguished elements, and the operation formulas from which the system is built.

Definition 0.18.3 (Construction Verification)**.** *Construction verification* is the proof that the construction data actually satisfies the construction target. It includes, as applicable, closure of operations, equivalence-relation proofs, well-definedness of operations on classes, algebraic laws, order laws, and compatibility of embeddings with earlier systems.

0.19 Finite Modular Number Systems

Congruence and Residue Classes

Definition 0.19.1 (Congruence Modulo an Integer)**.** Let $n \in \mathbb{Z}$ be positive. For $a, b \in \mathbb{Z}$, define

$$a \equiv b \pmod{n} \quad :\Longleftrightarrow \quad n \mid (a - b).$$

Theorem 0.19.2 (Congruence Modulo an Integer Is an Equivalence Relation)**.** *For each positive integer n, congruence modulo n is an equivalence relation on $\mathbb{Z}$.*

Definition 0.19.3 (Integers Modulo n)**.** Let $n \in \mathbb{Z}$ be positive. The *integers modulo n* are the quotient set

$$\mathbb{Z}/n\mathbb{Z} := \mathbb{Z}/\equiv \pmod{n}.$$

Definition 0.19.4 (Residue Class Modulo n)**.** For $a \in \mathbb{Z}$, the *residue class of a modulo n* is

$$[a]_n := \{b \in \mathbb{Z} : b \equiv a \pmod{n}\}.$$

Operations

Definition 0.19.5 (Addition Modulo n)**.** For residue classes modulo a positive integer n, define

$$[a]_n + [b]_n := [a + b]_n.$$

Definition 0.19.6 (Multiplication Modulo n)**.** For residue classes modulo a positive integer n, define

$$[a]_n \cdot [b]_n := [ab]_n.$$

Theorem 0.19.7 (Modular Addition and Multiplication Are Well-Defined)**.** *For each positive integer n, addition and multiplication modulo n are well-defined operations on $\mathbb{Z}/n\mathbb{Z}$.*

Definition 0.19.8 (Finite Modular Number System)**.** For a positive integer n, the *finite modular number system modulo n* is

$$\mathbb{Z}/n\mathbb{Z}$$

with addition and multiplication modulo n, additive identity $[0]_n$, and multiplicative identity $[1]_n$.

Basic Example

Theorem 0.19.9 ($\mathbb{Z}/4\mathbb{Z}$ Has Zero Divisors)**.** *In $\mathbb{Z}/4\mathbb{Z}$, the nonzero class $[2]_4$ satisfies*

$$[2]_4 \cdot [2]_4 = [0]_4.$$

Therefore $\mathbb{Z}/4\mathbb{Z}$ is not a field.

0.20 The $\{0, 1\}$ Number System

The Carrier and Operations

Definition 0.20.1 (Two-Element Carrier)**.** The *two-element carrier* is the set

$$\mathbb{B}_2 := \{0, 1\}.$$

Definition 0.20.2 (Addition on the Two-Element System)**.** Addition on $\mathbb{B}_2$ is the binary operation $+_2$ determined by

$$\begin{array}{c|cc} +_2 & 0 & 1 \\ \hline 0 & 0 & 1 \\ 1 & 1 & 0 \end{array}$$

Definition 0.20.3 (Multiplication on the Two-Element System)**.** Multiplication on $\mathbb{B}_2$ is the binary operation $\cdot_2$ determined by

$$\begin{array}{c|cc} \cdot_2 & 0 & 1 \\ \hline 0 & 0 & 0 \\ 1 & 0 & 1 \end{array}$$

Definition 0.20.4 (Two-Element Number System)**.** The *two-element number system* is

$$\mathbb{F}_2 := (\mathbb{B}_2, +_2, \cdot_2, 0, 1).$$

Verification

Theorem 0.20.5 (The Two-Element System Is a Field)**.** *The two-element number system $\mathbb{F}_2$ is a field.*

Theorem 0.20.6 (The Two-Element System Is Not an Ordered Field)**.** *There is no order on $\mathbb{B}_2$ that makes $\mathbb{F}_2$ an ordered field.*

As Modular Arithmetic

Theorem 0.20.7 ($\mathbb{Z}/2\mathbb{Z}$ Is the Two-Element System)**.** *The finite modular number system $\mathbb{Z}/2\mathbb{Z}$ has exactly two classes, $[0]_2$ and $[1]_2$, and its addition and multiplication tables are the tables of $\mathbb{F}_2$.*

Proofs

Capstone

Peano Systems

0.21 Presburger Arithmetic

Definition 0.21.1 (Presburger Signature)**.** The *Presburger signature* is the first-order signature

$$\mathcal{L}_{\text{Pres}} = \{0, S, +, <, =\},$$

where 0 is a constant symbol, S is a unary function symbol, $+$ is a binary function symbol, and $<$ and $=$ are binary relation symbols.

Definition 0.21.2 (Presburger Arithmetic)**.** *Presburger arithmetic* is the first-order theory of the natural numbers in the Presburger signature. Its intended structure is

$$(\mathbb{N}, 0, S, +, <, =),$$

and its axioms describe the behavior of zero, successor, addition, order, and induction for formulas written in the language $\mathcal{L}_{\text{Pres}}$.

0.22 Defining Peano Systems

Definition 0.22.1 (Peano System)**.** A triple $(P, S, 1)$ is called a *Peano system* exactly when P is a set, $1 \in P$, $S : P \to P$ is a function, and the following axioms hold:

P1. $1 \in P$.

P2. For every $n \in P$, $S(n) \in P$.

P3. For every $n \in P$, $S(n) \neq 1$.

P4. For all $m, n \in P$, if $S(m) = S(n)$ then $m = n$.

P5. If $A \subseteq P$, $1 \in A$, and

$$\forall n \in P \; (n \in A \Longrightarrow S(n) \in A),$$

then $A = P$.

First Theorems on Peano Systems

Theorem 0.22.2 (Induction Principle for a Peano System)**.** *Let $(P, S, 1)$ be a Peano system, and let φ be a predicate on P. If*

$$\varphi(1) \qquad \textit{and} \qquad \forall n \in P \; (\varphi(n) \Longrightarrow \varphi(S(n))),$$

then

$$\forall n \in P \; \varphi(n).$$

Go to proof.

Theorem 0.22.3 (Strong Induction on a Peano System)**.** *Let $(P, S, 1)$ be a Peano system, and let φ be a predicate on P. If*

$$\forall n \in P \; \big(\forall k \in P \; (k < n \Longrightarrow \varphi(k))\big) \Longrightarrow \varphi(n),$$

then

$$\forall n \in P \; \varphi(n).$$

Go to proof.

Definition 0.22.4 (Successor-Closed Subset)**.** Let $(P, S, 1)$ be a Peano system, and let $A \subseteq P$. The subset A is called *successor-closed* exactly when

$$\forall n \in P \; (n \in A \Longrightarrow S(n) \in A).$$

Definition 0.22.5 (Inductive Subset of a Peano System)**.** Let $(P, S, 1)$ be a Peano system, and let $A \subseteq P$. The subset A is called *inductive* exactly when $1 \in A$ and A is successor-closed.

Theorem 0.22.6 (Minimality of a Peano System)**.** *Let $(P, S, 1)$ be a Peano system. If $A \subseteq P$ contains 1 and is closed under successor, then $A = P$.*

$$\big(1 \in A \land \forall n \in P \; (n \in A \Longrightarrow S(n) \in A)\big) \Longrightarrow A = P.$$

Go to proof.

Definition 0.22.7 (Predecessor in a Peano System)**.** Let $(P, S, 1)$ be a Peano system, and let $x, u \in P$. The element u is called a *predecessor* of x exactly when

$$S(u) = x.$$

Theorem 0.22.8 (Every Element Is Either 1 or a Successor)**.** *Let $(P, S, 1)$ be a Peano system. For every $x \in P$, either $x = 1$ or there exists $u \in P$ such that $S(u) = x$. Go to proof.*

Theorem 0.22.9 (Successor Inequality Reflection)**.** *Let $(P, S, 1)$ be a Peano system. For all $a, b \in P$, if $a \neq b$, then $S(a) \neq S(b)$. Go to proof.*

Corollary 0.22.10 (Non-One Elements Have a Predecessor)**.** *Let $(P, S, 1)$ be a Peano system. For every $x \in P$, if $x \neq 1$, then there exists $u \in P$ such that $S(u) = x$. Go to proof.*

Corollary 0.22.11 (Predecessor Existence and Uniqueness Away from 1)**.** *Let $(P, S, 1)$ be a Peano system, and let $x \in P$ with $x \neq 1$. Then there exists a unique $u \in P$ such that $S(u) = x$. Go to proof.*

Corollary 0.22.12 (Unique Predecessor Characterization Away from 1)**.** *Let $(P, S, 1)$ be a Peano system. For every $x \in P$, $x \neq 1$ if and only if there exists a unique $u \in P$ such that $S(u) = x$. Go to proof.*

Corollary 0.22.13 (The Distinguished Element Is the Unique Non-Successor)**.** *Let $(P, S, 1)$ be a Peano system. An element $x \in P$ is not a successor of any element of P if and only if $x = 1$. Go to proof.*

Theorem 0.22.14 (No Object Is Its Own Successor)**.** *Let $(P, S, 1)$ be a Peano system. For every $n \in P$,*

$$S(n) \neq n.$$

Go to proof.

Peano System Figure Plates

0.23 Examples of Peano Systems

The Standard One-Based System

Von Neumann Ordinals

Nonnumeric Peano Systems

A Finite Alphabet Is Not a Peano System

Summary

0.24 Recursion on Peano Systems

Arity Examples Before the Theorem

Formal Development

Definition 0.24.1 (Iterator Data)**.** Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $g : W \to W$ be a function. The triple (W, c, g) is called *iterator data* for $(P, S, 1)$.

Definition 0.24.2 (Iterator Relation)**.** Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. A relation $R \subseteq P \times W$ is called an *iterator relation* for (W, c, g) exactly when

$$(1, c) \in R$$

and

$$\forall n \in P \, \forall w \in W \, \big((n, w) \in R \Longrightarrow (S(n), g(w)) \in R\big).$$

Definition 0.24.3 (Minimal Iterator Relation)**.** Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. The *minimal iterator relation* for (W, c, g) is

$$R^* = \bigcap \{R \subseteq P \times W : R \text{ is an iterator relation for } (W, c, g)\}.$$

Lemma 0.24.4 (Consistency of the Minimal Iterator Relation)**.** *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation for (W, c, g). Then R^* is an iterator relation for (W, c, g). Go to proof.*

Lemma 0.24.5 (Forced Values Are Unique)**.** *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation. If $(n, w_0) \in R^*$ and $(n, w_1) \in R^*$, then $w_0 = w_1$. Go to proof.*

Lemma 0.24.6 (Determinism of the Minimal Iterator Relation)**.** *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation. For each $n \in P$, there is at most one $w \in W$ such that $(n, w) \in R^*$. Go to proof.*

Lemma 0.24.7 (Completeness of the Minimal Iterator Relation)**.** *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation. For each $n \in P$, there exists $w \in W$ such that $(n, w) \in R^*$. Go to proof.*

Lemma 0.24.8 (Uniqueness of Iterator Functions)**.** *Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. If $f, h : P \to W$ satisfy*

$$f(1) = c, \qquad h(1) = c,$$

and

$$\forall n \in P \; (f(S(n)) = g(f(n))) \qquad \text{and} \qquad \forall n \in P \; (h(S(n)) = g(h(n))),$$

then $f = h$. Go to proof.

Lemma 0.24.9 (Existence of an Iterator Function)**.** *Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. Then there exists a function $f : P \to W$ such that*

$$f(1) = c \qquad \textit{and} \qquad \forall n \in P\ (f(S(n)) = g(f(n))).$$

Go to proof.

Theorem 0.24.10 (Peano Iterator Theorem)**.** *Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. Then there exists a unique function $f : P \to W$ such that*

$$f(1) = c \qquad \textit{and} \qquad \forall n \in P\ (f(S(n)) = g(f(n))).$$

Go to proof.

Definition 0.24.11 (Iterator-Generated Function)**.** Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. The unique function $f : P \to W$ satisfying

$$f(1) = c \qquad \text{and} \qquad \forall n \in P\ (f(S(n)) = g(f(n)))$$

is called the *iterator-generated function determined by* (W, c, g).

Theorem 0.24.12 (Iterator Base Clause)**.** *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let f be the iterator-generated function determined by (W, c, g). Then*

$$f(1) = c.$$

Go to proof.

Theorem 0.24.13 (Iterator Successor Clause)**.** *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let f be the iterator-generated function determined by (W, c, g). Then*

$$\forall n \in P\ (f(S(n)) = g(f(n))).$$

Go to proof.

Definition 0.24.14 (Iteration of a Self-Map)**.** Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $g : W \to W$. The iterator-generated function determined by (W, c, g) is called the *iteration of g from c along P*.

Corollary 0.24.15 (Uniqueness of Binary Iterator Operations)**.** *Let $(P, S, 1)$ be a Peano system, let A be a set, let W be a set, and suppose each $a \in A$ determines iterator data (W, c_a, g_a) for $(P, S, 1)$. If $F, G : A \times P \to W$ satisfy*

$$F(a, 1) = c_a, \qquad G(a, 1) = c_a$$

and

$$F(a, S(n)) = g_a(F(a, n)), \qquad G(a, S(n)) = g_a(G(a, n))$$

for all $a \in A$ and all $n \in P$, then $F = G$. *Go to proof.*

Definition 0.24.16 (Stage-Dependent Step Rule)**.** Let $(P, S, 1)$ be a Peano system and let W be a set. A *stage-dependent step rule* on W over P is a function

$$\Phi : P \times W \to W.$$

Definition 0.24.17 (General Recursive Function)**.** Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. A function $f : P \to W$ is called the *general recursive function determined by* (W, c, Φ) exactly when

$$f(1) = c \qquad \text{and} \qquad \forall n \in P\ (f(S(n)) = \Phi(n, f(n))).$$

Lemma 0.24.18 (Uniqueness of General Recursive Functions)**.** *Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. If $f, h : P \to W$ satisfy*

$$f(1) = c, \qquad h(1) = c,$$

and

$$\forall n \in P\ (f(S(n)) = \Phi(n, f(n))) \qquad \text{and} \qquad \forall n \in P\ (h(S(n)) = \Phi(n, h(n))),$$

then $f = h$. *Go to proof.*

Theorem 0.24.19 (General Recursion by State Encoding)**.** *Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. There exists a function $H : P \to P \times W$ such that*

$$H(1) = (1, c)$$

and

$$\forall n \in P\ \big(H(S(n)) = (S(\pi_1(H(n))), \Phi(\pi_1(H(n)), \pi_2(H(n))))\big).$$

Go to proof.

Theorem 0.24.20 (General Recursion Theorem)**.** *Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. Then there exists a unique function $f : P \to W$ such that*

$$f(1) = c \qquad \text{and} \qquad \forall n \in P\ (f(S(n)) = \Phi(n, f(n))).$$

Go to proof.

Theorem 0.24.21 (Uniqueness of Peano Systems up to Isomorphism)**.** *Let $(P, S_P, 1_P)$ and $(Q, S_Q, 1_Q)$ be Peano systems. Then there exists a unique bijection $\theta : P \to Q$ such that*

$$\theta(1_P) = 1_Q$$

and

$$\forall n \in P\ (\theta(S_P(n)) = S_Q(\theta(n))).$$

Thus any two Peano systems are isomorphic. *Go to proof.*

Proofs

Capstone

Natural Numbers ($\mathbb{N}$)

0.25 Axioms for $\mathbb{N}$

0.25.1 Axioms for $\mathbb{N}$

Definition 0.25.1 (The Natural Numbers)**.** The natural numbers $\mathbb{N}$ are the canonical Peano system

$$(\mathbb{N}, 1, S),$$

where $1 \in \mathbb{N}$ is the distinguished base element and $S : \mathbb{N} \to \mathbb{N}$ is the successor operation.

0.26 Canonical Natural Number System

0.27 Addition on N

Definition 0.27.1 (Addition on a Peano System)**.** Let $(P, S, 1)$ be a Peano system. For each $m \in P$, define the function

$$f_m : P \to P$$

to be the unique function satisfying

$$f_m(1) = S(m) \qquad \text{and} \qquad \forall n \in P \ (f_m(S(n)) = S(f_m(n))).$$

For $m, n \in P$, define

$$m + n := f_m(n).$$

Theorem 0.27.2 (Addition Is Well-Defined on a Peano System)**.** *Let $(P, S, 1)$ be a Peano system. For every $m \in P$, there exists a unique function*

$$f_m : P \to P$$

such that

$$f_m(1) = S(m) \qquad \textit{and} \qquad \forall n \in P \ (f_m(S(n)) = S(f_m(n))).$$

Consequently the rule

$$(m, n) \longmapsto m + n := f_m(n)$$

defines a unique binary operation on P. *Go to proof.*

Theorem 0.27.3 (Addition With 1)**.** *Let $(P, S, 1)$ be a Peano system. For every $m \in P$,*

$$m + 1 = S(m).$$

Go to proof.

Theorem 0.27.4 (Addition Successor on the Right)**.** *Let $(P, S, 1)$ be a Peano system. For every $m, n \in P$,*

$$m + S(n) = S(m + n).$$

Go to proof.

Theorem 0.27.5 (One Plus n). *Let $(P, S, 1)$ be a Peano system. For every $n \in P$,*

$$1 + n = S(n).$$

Go to proof.

Theorem 0.27.6 (Addition Is Associative). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$(a + b) + c = a + (b + c).$$

Go to proof.

Theorem 0.27.7 (Addition Is Commutative). *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$a + b = b + a.$$

Go to proof.

Theorem 0.27.8 (Left Cancellation for Addition). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a + b = a + c \Longrightarrow b = c.$$

Go to proof.

Theorem 0.27.9 (Right Cancellation for Addition). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$b + a = c + a \Longrightarrow b = c.$$

Go to proof.

Theorem 0.27.10 (Cancellation for Addition). *Let $(P, S, 1)$ be a Peano system. Addition on P satisfies cancellation on both sides:*

$$a + b = a + c \Longrightarrow b = c \qquad \text{and} \qquad b + a = c + a \Longrightarrow b = c.$$

Go to proof.

Theorem 0.27.11 (No Natural Number Equals Itself Plus a Natural Number). *Let $(P, S, 1)$ be a Peano system. For every $m, n \in P$,*

$$n \neq m + n.$$

Go to proof.

0.28 Multiplication on N

Definition 0.28.1 (Multiplication on a Peano System). Let $(P, S, 1)$ be a Peano system. For each $m \in P$, define the function

$$g_m : P \to P$$

to be the unique function satisfying

$$g_m(1) = m \qquad \text{and} \qquad \forall n \in P \; (g_m(S(n)) = g_m(n) + m).$$

For $m, n \in P$, define

$$m \cdot n \coloneqq g_m(n).$$

Theorem 0.28.2 (Multiplication Is Well-Defined on a Peano System). *Let $(P, S, 1)$ be a Peano system. For every $m \in P$, there exists a unique function*

$$g_m : P \to P$$

such that

$$g_m(1) = m \qquad \text{and} \qquad \forall n \in P \; (g_m(S(n)) = g_m(n) + m).$$

Consequently the rule

$$(m, n) \longmapsto m \cdot n \coloneqq g_m(n)$$

defines a unique binary operation on P. Go to proof.

Theorem 0.28.3 (Multiplication With 1)**.** *Let $(P, S, 1)$ be a Peano system. For every $m \in P$,*

$$m \cdot 1 = m.$$

Go to proof.

Theorem 0.28.4 (Multiplication Successor on the Right)**.** *Let $(P, S, 1)$ be a Peano system. For every $m, n \in P$,*

$$m \cdot S(n) = (m \cdot n) + m.$$

Go to proof.

Theorem 0.28.5 (One Times n)**.** *Let $(P, S, 1)$ be a Peano system. For every $n \in P$,*

$$1 \cdot n = n.$$

Go to proof.

Theorem 0.28.6 (Multiplication Distributes Over Addition)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c).$$

Go to proof.

Theorem 0.28.7 (Left Distributivity of Multiplication Over Addition)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$(a + b) \cdot c = (a \cdot c) + (b \cdot c).$$

Go to proof.

Theorem 0.28.8 (Multiplication Distributes over Addition on Both Sides)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c) \qquad \textit{and} \qquad (a + b) \cdot c = (a \cdot c) + (b \cdot c).$$

Go to proof.

Theorem 0.28.9 (Multiplication Is Associative)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$(a \cdot b) \cdot c = a \cdot (b \cdot c).$$

Go to proof.

Theorem 0.28.10 (Multiplication Is Commutative)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$a \cdot b = b \cdot a.$$

Go to proof.

Theorem 0.28.11 (Left Cancellation for Multiplication)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a \cdot b = a \cdot c \Longrightarrow b = c.$$

Go to proof.

Theorem 0.28.12 (Right Cancellation for Multiplication)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$b \cdot a = c \cdot a \Longrightarrow b = c.$$

Go to proof.

Theorem 0.28.13 (Cancellation for Multiplication)**.** *Let $(P, S, 1)$ be a Peano system. Multiplication on P satisfies cancellation on both sides:*

$$a \cdot b = a \cdot c \Longrightarrow b = c \qquad \textit{and} \qquad b \cdot a = c \cdot a \Longrightarrow b = c.$$

Go to proof.

0.29 Order on N

Definition 0.29.1 (Strict Less Than on a Peano System)**.** Let $(P, S, 1)$ be a Peano system, and let $a, b \in P$. One writes

$$a < b$$

exactly when there exists $c \in P$ such that

$$b = a + c.$$

Definition 0.29.2 (Less Than or Equal on a Peano System)**.** Let $(P, S, 1)$ be a Peano system, and let $a, b \in P$. One writes

$$a \leq b$$

exactly when

$$a < b \qquad \text{or} \qquad a = b.$$

Theorem 0.29.3 (Order Is Reflexive on a Peano System)**.** *Let $(P, S, 1)$ be a Peano system. For every $a \in P$,*

$$a \leq a.$$

Go to proof.

Theorem 0.29.4 (Order Is Transitive on a Peano System)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$\big(a \leq b \wedge b \leq c\big) \Longrightarrow a \leq c.$$

Go to proof.

Theorem 0.29.5 (Order Is Antisymmetric on a Peano System)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$\big(a \leq b \wedge b \leq a\big) \Longrightarrow a = b.$$

Go to proof.

Theorem 0.29.6 (Addition Preserves Order on the Right)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a \leq b \Longrightarrow a + c \leq b + c.$$

Go to proof.

Theorem 0.29.7 (Addition Preserves Order on the Left)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a \leq b \Longrightarrow c + a \leq c + b.$$

Go to proof.

Theorem 0.29.8 (Addition Preserves Order)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a \leq b \Longrightarrow a + c \leq b + c \qquad \textit{and} \qquad a \leq b \Longrightarrow c + a \leq c + b.$$

Go to proof.

Theorem 0.29.9 (Multiplication Preserves and Reflects Strict Order)**.** *Let $(P, S, 1)$ be a Peano system. For every $k, m, n \in P$,*

$$m < n \Longleftrightarrow k \cdot m < k \cdot n.$$

Go to proof.

Theorem 0.29.10 (Strict Order Successor Characterization)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$a < b \Longleftrightarrow S(a) \leq b.$$

Go to proof.

Theorem 0.29.11 (Trichotomy on a Peano System). *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$, exactly one of the following holds:*

$$a < b, \qquad a = b, \qquad b < a.$$

Go to proof.

Theorem 0.29.12 (Natural-Number Order Is Total). *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$a \leq b \qquad \text{or} \qquad b \leq a.$$

Consequently $\leq$ is a total order on P. *Go to proof.*

0.30 Powers and Growth

Definition 0.30.1 (Exponentiation on a Peano System). Let $(P, S, 1)$ be a Peano system. For each $a \in P$, define the function

$$h_a : P \to P$$

to be the unique function satisfying

$$h_a(1) = a \qquad \text{and} \qquad \forall n \in P \ (h_a(S(n)) = h_a(n) \cdot a).$$

For $a, n \in P$, define

$$a^n := h_a(n).$$

Theorem 0.30.2 (Exponentiation Is Well-Defined on a Peano System). *Let $(P, S, 1)$ be a Peano system. For every $a \in P$, there exists a unique function*

$$h_a : P \to P$$

such that

$$h_a(1) = a \qquad \text{and} \qquad \forall n \in P \ (h_a(S(n)) = h_a(n) \cdot a).$$

Consequently the rule

$$(a, n) \longmapsto a^n := h_a(n)$$

defines a unique binary operation on P. *Go to proof.*

Theorem 0.30.3 (Exponent Sum Law). *Let $(P, S, 1)$ be a Peano system. For all $a, m, n \in P$,*

$$a^{m+n} = a^m \cdot a^n.$$

Go to proof.

Theorem 0.30.4 (Exponent Product Law). *Let $(P, S, 1)$ be a Peano system. For all $a, b, n \in P$,*

$$(a \cdot b)^n = a^n \cdot b^n.$$

Go to proof.

Theorem 0.30.5 (Exponent Power Law). *Let $(P, S, 1)$ be a Peano system. For all $a, m, n \in P$,*

$$(a^m)^n = a^{m \cdot n}.$$

Go to proof.

Theorem 0.30.6 (Powers with Common Exponent Preserve Strict Order). *Let $(P, S, 1)$ be a Peano system. For every $a, b, n \in P$,*

$$a < b \iff a^n < b^n.$$

Go to proof.

Theorem 0.30.7 (Powers with Common Base Preserve Strict Order)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, m, n \in P$,*

$$a^m < a^n \iff 1 < a \wedge m < n.$$

Go to proof.

Theorem 0.30.8 ($n < 2^n$ for All $n \in \mathbb{N}$)**.** *Let $(P, S, 1)$ be a Peano system, and let $2 := S(1)$. For every $n \in P$,*

$$n < 2^n.$$

Go to proof.

0.31 Well Ordering Principle

Theorem 0.31.1 (Well-Ordering Principle)**.** *Let $(P, S, 1)$ be a Peano system. Every nonempty subset of P has a least element with respect to the order $\leq$ on P. Go to proof.*

Theorem 0.31.2 (Induction, Strong Induction, and Well-Ordering Are Equivalent)**.** *Let $(P, S, 1)$ be a Peano system equipped with the order $\leq$. The following are equivalent:*

1. *The induction principle: for every predicate φ on P, if $\varphi(1)$ holds and the successor step holds, then $\varphi(n)$ holds for all $n \in P$.*
2. *The strong induction principle: for every predicate φ on P, if $\varphi(k)$ holds for every $k < n$ implies $\varphi(n)$, then $\varphi(n)$ holds for every $n \in P$.*
3. *The well-ordering principle: every nonempty subset of P has a least element with respect to $\leq$.*

Go to proof.

0.32 Induction

Theorem 0.32.1 (Induction from an Arbitrary Base)**.** *Let $(P, S, 1)$ be a Peano system, let $m \in P$, and let φ be a predicate on P. If*

$$\varphi(m) \qquad \text{and} \qquad \forall n \in P \ \big(n \geq m \wedge \varphi(n) \implies \varphi(S(n))\big),$$

then

$$\forall n \in P \ \big(n \geq m \implies \varphi(n)\big).$$

Go to proof.

0.33 Algebraic Structure of N

Theorem 0.33.1 ($\mathbb{N}$ Is Closed, Associative, and Commutative Under Addition)**.** *Let $(P, S, 1)$ be a Peano system. The binary operation $+$ on P satisfies:*

1. ***Closure.*** *For all $a, b \in P$, $a + b \in P$.*
2. ***Associativity.*** *For all $a, b, c \in P$, $(a + b) + c = a + (b + c)$.*
3. ***Commutativity.*** *For all $a, b \in P$, $a + b = b + a$.*
4. ***Left cancellation.*** *For all $a, b, c \in P$, if $a + b = a + c$ then $b = c$.*
5. ***Right cancellation.*** *For all $a, b, c \in P$, if $b + a = c + a$ then $b = c$.*

Go to proof.

Theorem 0.33.2 ($\mathbb{N}$ Is a Commutative Monoid Under Multiplication)**.** *Let $(P, S, 1)$ be a Peano system. The binary operation $\cdot$ on P satisfies:*

1. ***Closure.*** *For all $a, b \in P$, $a \cdot b \in P$.*
2. ***Associativity.*** *For all $a, b, c \in P$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.*
3. ***Commutativity.*** *For all $a, b \in P$, $a \cdot b = b \cdot a$.*
4. ***Identity.*** *For all $a \in P$, $1 \cdot a = a = a \cdot 1$.*

Consequently $(P, \cdot, 1)$ is a commutative monoid. Go to proof.

Theorem 0.33.3 ($\mathbb{N}$ Is a Semiring)**.** *Let $(P, S, 1)$ be a Peano system. Then $(P, +, \cdot)$ satisfies:*

1. *$(P, +)$ is a commutative semigroup.*
2. *$(P, \cdot, 1)$ is a commutative monoid.*
3. *Multiplication distributes over addition: for all $a, b, c \in P$,*

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c).$$

Go to proof.

Proofs

Capstone

Constructing the Whole Numbers

0.34 Why Whole Numbers Are Introduced

The Missing Identity

0.35 Definition of the Whole Numbers

The Underlying Set

Definition 0.35.1 (Whole Numbers)**.** Let $\mathbb{N}$ be the one-based natural-number system developed in the preceding chapter. Let 0 be a new element with $0 \notin \mathbb{N}$. Define

$$\mathbb{W} \coloneqq \mathbb{N} \cup \{0\}.$$

Theorem 0.35.2 ($0 \notin \mathbb{N}$)**.** *The adjoined element* 0 *is not an element of the one-based natural-number system:*

$$0 \notin \mathbb{N}.$$

Go to proof.

0.36 Extending Successor

The Extended Successor

Definition 0.36.1 (Successor on $\mathbb{W}$)**.** Define $S_{\mathbb{W}} : \mathbb{W} \to \mathbb{W}$ by

$$S_{\mathbb{W}}(0) = 1 \qquad \text{and} \qquad S_{\mathbb{W}}(n) = S_{\mathbb{N}}(n) \quad (n \in \mathbb{N}).$$

Theorem 0.36.2 (Successor on $\mathbb{W}$ Is Well-Defined)**.** *The rule defining $S_{\mathbb{W}}$ determines a function $S_{\mathbb{W}} : \mathbb{W} \to \mathbb{W}$.* *Go to proof.*

Theorem 0.36.3 (0 Is Not a Successor in $\mathbb{W}$)**.** *There is no $a \in \mathbb{W}$ such that $S_{\mathbb{W}}(a) = 0$.* *Go to proof.*

Theorem 0.36.4 (Every Nonzero Element of $\mathbb{W}$ Is a Successor)**.** *If $a \in \mathbb{W}$ and $a \neq 0$, then there exists $b \in \mathbb{W}$ such that $a = S_{\mathbb{W}}(b)$.* *Go to proof.*

Theorem 0.36.5 (Successor on $\mathbb{W}$ Is Injective)**.** *For all $a, b \in \mathbb{W}$,*

$$S_{\mathbb{W}}(a) = S_{\mathbb{W}}(b) \implies a = b.$$

Go to proof.

Theorem 0.36.6 (($\mathbb{W}, 0, S_{\mathbb{W}}$) Satisfies the Successor Properties)**.** *The triple* $(\mathbb{W}, 0, S_{\mathbb{W}})$ *satisfies the successor-side properties:*

$$0 \in \mathbb{W}, \qquad S_{\mathbb{W}} : \mathbb{W} \to \mathbb{W},$$
$$\neg \exists a \in \mathbb{W}\ (S_{\mathbb{W}}(a) = 0), \qquad \forall a, b \in \mathbb{W}\ (S_{\mathbb{W}}(a) = S_{\mathbb{W}}(b) \implies a = b).$$

Go to proof.

0.37 Extending Addition to $\mathbb{W}$

0.37.1 Extending Addition to $\mathbb{W}$

The Extended Addition Operation

Definition 0.37.1 (Addition on $\mathbb{W}$)**.** Define $+_{\mathbb{W}} : \mathbb{W} \times \mathbb{W} \to \mathbb{W}$ by

$$0 +_{\mathbb{W}} a = a, \qquad a +_{\mathbb{W}} 0 = a,$$

and, for $a, b \in \mathbb{N}$,

$$a +_{\mathbb{W}} b = a +_{\mathbb{N}} b.$$

Theorem 0.37.2 (Addition on $\mathbb{W}$ Is Well-Defined)**.** *The case definition of $+_{\mathbb{W}}$ determines a binary operation on $\mathbb{W}$. Go to proof.*

Theorem 0.37.3 (Zero Is an Additive Identity on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$0 +_{\mathbb{W}} a = a \qquad \textit{and} \qquad a +_{\mathbb{W}} 0 = a.$$

Go to proof.

Theorem 0.37.4 (Left Additive Identity on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$0 +_{\mathbb{W}} a = a.$$

Go to proof.

Theorem 0.37.5 (Right Additive Identity on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$a +_{\mathbb{W}} 0 = a.$$

Go to proof.

Theorem 0.37.6 (Addition on $\mathbb{W}$ Extends Addition on $\mathbb{N}$)**.** *For all $a, b \in \mathbb{N}$,*

$$a +_{\mathbb{W}} b = a +_{\mathbb{N}} b.$$

Go to proof.

Theorem 0.37.7 (Addition on $\mathbb{W}$ Is Associative)**.** *For all $a, b, c \in \mathbb{W}$,*

$$(a +_{\mathbb{W}} b) +_{\mathbb{W}} c = a +_{\mathbb{W}} (b +_{\mathbb{W}} c).$$

Go to proof.

Theorem 0.37.8 (Addition on $\mathbb{W}$ Is Commutative)**.** *For all $a, b \in \mathbb{W}$,*

$$a +_{\mathbb{W}} b = b +_{\mathbb{W}} a.$$

Go to proof.

Theorem 0.37.9 (Addition on $\mathbb{W}$ Satisfies Cancellation)**.** *For all $a, b, c \in \mathbb{W}$,*

$$a +_{\mathbb{W}} b = a +_{\mathbb{W}} c \Longrightarrow b = c.$$

Go to proof.

0.38 Extending Multiplication to $\mathbb{W}$

0.38.1 Extending Multiplication to $\mathbb{W}$

The Extended Multiplication Operation

Definition 0.38.1 (Multiplication on $\mathbb{W}$)**.** Define $\cdot_{\mathbb{W}} : \mathbb{W} \times \mathbb{W} \to \mathbb{W}$ by

$$0 \cdot_{\mathbb{W}} a = 0, \qquad a \cdot_{\mathbb{W}} 0 = 0,$$

and, for $a, b \in \mathbb{N}$,

$$a \cdot_{\mathbb{W}} b = a \cdot_{\mathbb{N}} b.$$

Theorem 0.38.2 (Multiplication on $\mathbb{W}$ Is Well-Defined)**.** *The case definition of $\cdot_{\mathbb{W}}$ determines a binary operation on $\mathbb{W}$. Go to proof.*

Theorem 0.38.3 (Zero Is Absorbing for Multiplication on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$0 \cdot_{\mathbb{W}} a = 0 \qquad \textit{and} \qquad a \cdot_{\mathbb{W}} 0 = 0.$$

Go to proof.

Theorem 0.38.4 (Left Zero Absorption on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$0 \cdot_{\mathbb{W}} a = 0.$$

Go to proof.

Theorem 0.38.5 (Right Zero Absorption on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$a \cdot_{\mathbb{W}} 0 = 0.$$

Go to proof.

Theorem 0.38.6 (One Is a Multiplicative Identity on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$1 \cdot_{\mathbb{W}} a = a \qquad \textit{and} \qquad a \cdot_{\mathbb{W}} 1 = a.$$

Go to proof.

Theorem 0.38.7 (Multiplication on $\mathbb{W}$ Extends Multiplication on $\mathbb{N}$)**.** *For all $a, b \in \mathbb{N}$,*

$$a \cdot_{\mathbb{W}} b = a \cdot_{\mathbb{N}} b.$$

Go to proof.

Theorem 0.38.8 (Multiplication on $\mathbb{W}$ Is Associative)**.** *For all $a, b, c \in \mathbb{W}$,*

$$(a \cdot_{\mathbb{W}} b) \cdot_{\mathbb{W}} c = a \cdot_{\mathbb{W}} (b \cdot_{\mathbb{W}} c).$$

Go to proof.

Theorem 0.38.9 (Multiplication on $\mathbb{W}$ Is Commutative)**.** *For all $a, b \in \mathbb{W}$,*

$$a \cdot_{\mathbb{W}} b = b \cdot_{\mathbb{W}} a.$$

Go to proof.

Theorem 0.38.10 (Multiplication Distributes over Addition on $\mathbb{W}$)**.** *For all $a, b, c \in \mathbb{W}$,*

$$a \cdot_{\mathbb{W}} (b +_{\mathbb{W}} c) = (a \cdot_{\mathbb{W}} b) +_{\mathbb{W}} (a \cdot_{\mathbb{W}} c).$$

Go to proof.

0.39 Order on $\mathbb{W}$

0.39.1 Order on $\mathbb{W}$

The Extended Order

Definition 0.39.1 (Order on $\mathbb{W}$)**.** Define $\leq_{\mathbb{W}}$ on $\mathbb{W}$ by declaring

$$0 \leq_{\mathbb{W}} a \quad (a \in \mathbb{W}),$$

and, for $a, b \in \mathbb{N}$,

$$a \leq_{\mathbb{W}} b \quad \Longleftrightarrow \quad a \leq_{\mathbb{N}} b.$$

Definition 0.39.2 (Strict Order on $\mathbb{W}$)**.** Define $<_{\mathbb{W}}$ on $\mathbb{W}$ by declaring

$$0 <_{\mathbb{W}} n \quad (n \in \mathbb{N}),$$

and, for $a, b \in \mathbb{N}$,

$$a <_{\mathbb{W}} b \quad \Longleftrightarrow \quad a <_{\mathbb{N}} b.$$

No element of $\mathbb{W}$ is strictly less than 0.

Theorem 0.39.3 (Order on $\mathbb{W}$ Extends Order on $\mathbb{N}$)**.** *For all $a, b \in \mathbb{N}$,*

$$a \leq_{\mathbb{W}} b \Longleftrightarrow a \leq_{\mathbb{N}} b, \qquad a <_{\mathbb{W}} b \Longleftrightarrow a <_{\mathbb{N}} b.$$

Go to proof.

Theorem 0.39.4 (Zero Is the Least Element of $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$0 \leq_{\mathbb{W}} a.$$

Go to proof.

Theorem 0.39.5 (Order on $\mathbb{W}$ Is Reflexive)**.** *For every $a \in \mathbb{W}$, $a \leq_{\mathbb{W}} a$.* *Go to proof.*

Theorem 0.39.6 (Order on $\mathbb{W}$ Is Antisymmetric)**.** *For all $a, b \in \mathbb{W}$,*

$$a \leq_{\mathbb{W}} b \text{ and } b \leq_{\mathbb{W}} a \Longrightarrow a = b.$$

Go to proof.

Theorem 0.39.7 (Order on $\mathbb{W}$ Is Transitive)**.** *For all $a, b, c \in \mathbb{W}$,*

$$a \leq_{\mathbb{W}} b \text{ and } b \leq_{\mathbb{W}} c \Longrightarrow a \leq_{\mathbb{W}} c.$$

Go to proof.

Theorem 0.39.8 (Order on $\mathbb{W}$ Is Total)**.** *For all $a, b \in \mathbb{W}$,*

$$a \leq_{\mathbb{W}} b \quad \text{or} \quad b \leq_{\mathbb{W}} a.$$

Go to proof.

Theorem 0.39.9 (Well-Ordering Principle for $\mathbb{W}$)**.** *Every nonempty subset of $\mathbb{W}$ has a least element with respect to $\leq_{\mathbb{W}}$.* *Go to proof.*

0.40 Algebraic Structure of $\mathbb{W}$

0.40.1 Algebraic Structure of $\mathbb{W}$

Algebraic Packages

Theorem 0.40.1 (($\mathbb{W}, +, 0$) Is a Commutative Monoid)**.** *The structure* $(\mathbb{W}, +_{\mathbb{W}}, 0)$ *is a commutative monoid. Go to proof.*

Theorem 0.40.2 (($\mathbb{W}, \cdot, 1$) Is a Commutative Monoid)**.** *The structure* $(\mathbb{W}, \cdot_{\mathbb{W}}, 1)$ *is a commutative monoid. Go to proof.*

Theorem 0.40.3 (($\mathbb{W}, +, \cdot, 0, 1$) Is a Commutative Semiring)**.** *The structure* $(\mathbb{W}, +_{\mathbb{W}}, \cdot_{\mathbb{W}}, 0, 1)$ *is a commutative semiring. Go to proof.*

Theorem 0.40.4 ($\mathbb{W}$ Is a Commutative Ordered Semiring)**.** *With the operations* $+_{\mathbb{W}}$ *and* $\cdot_{\mathbb{W}}$ *and the order* $\leq_{\mathbb{W}}$*, the whole numbers form a commutative ordered semiring. Go to proof.*

Theorem 0.40.5 ($\mathbb{N}$ Embeds into $\mathbb{W}$ as the Positive Part)**.** *The inclusion map identifies* $\mathbb{N}$ *with the positive part* $\mathbb{W} \setminus \{0\}$ *and preserves successor,* 1*, addition, multiplication, and order. Go to proof.*

Theorem 0.40.6 ($\mathbb{W}$ Has No Additive Inverses Except 0)**.** *If* $a, b \in \mathbb{W}$ *and*

$$a +_{\mathbb{W}} b = 0,$$

then $a = 0$ *and* $b = 0$*. Go to proof.*

Theorem 0.40.7 ($\mathbb{W}$ Is Not a Ring)**.** *With the operations* $+_{\mathbb{W}}$ *and* $\cdot_{\mathbb{W}}$*, the whole numbers do not form a ring. Go to proof.*

0.41 Transition to $\mathbb{Z}$

0.41.1 Transition to $\mathbb{Z}$

The Next Construction

Proofs

Capstone

Integers ($\mathbb{Z}$)

0.42 Construction of $\mathbb{Z}$

Integer Prepairs

Definition 0.42.1 (Integer Prepair)**.** An *integer prepair* is an ordered pair $(a, b) \in \mathbb{W} \times \mathbb{W}$, read as the formal difference $a - b$.

Definition 0.42.2 (Integer Prepair Set)**.** The integer prepair set is

$$\operatorname{Pre}\mathbb{Z} := \mathbb{W} \times \mathbb{W}.$$

Formal-Difference Equivalence

Definition 0.42.3 (Formal-Difference Equivalence)**.** For $(a, b), (c, d) \in \operatorname{Pre}\mathbb{Z}$, define

$$(a, b) \sim_{\mathbb{Z}} (c, d) \quad :\Longleftrightarrow \quad a + d = c + b.$$

Lemma 0.42.4 (Formal-Difference Equivalence Is Reflexive)**.** *For every* $(a, b) \in \operatorname{Pre}\mathbb{Z}$*,*

$$(a, b) \sim_{\mathbb{Z}} (a, b).$$

Lemma 0.42.5 (Formal-Difference Equivalence Is Symmetric)**.** *For all prepairs* $(a, b), (c, d)$*,*

$$(a, b) \sim_{\mathbb{Z}} (c, d) \Longrightarrow (c, d) \sim_{\mathbb{Z}} (a, b).$$

Lemma 0.42.6 (Formal-Difference Equivalence Is Transitive)**.** *For all prepairs* $(a, b), (c, d), (e, f)$*,*

$$(a, b) \sim_{\mathbb{Z}} (c, d) \wedge (c, d) \sim_{\mathbb{Z}} (e, f) \Longrightarrow (a, b) \sim_{\mathbb{Z}} (e, f).$$

Theorem 0.42.7 (Formal-Difference Equivalence Is an Equivalence Relation)**.** *The relation* $\sim_{\mathbb{Z}}$ *is an equivalence relation on* $\operatorname{Pre}\mathbb{Z}$*.*

Definition of $\mathbb{Z}$

Definition 0.42.8 (Integers)**.** The integers are the quotient

$$\mathbb{Z} := \operatorname{Pre}\mathbb{Z}/{\sim_{\mathbb{Z}}}.$$

Definition 0.42.9 (Integer Class Notation)**.** The notation $[a, b]$ denotes the $\sim_{\mathbb{Z}}$-equivalence class of the prepair (a, b).

Zero and One in $\mathbb{Z}$

Definition 0.42.10 (Zero in $\mathbb{Z}$)**.** Define

$$0_{\mathbb{Z}} := [0, 0].$$

Definition 0.42.11 (One in $\mathbb{Z}$). Define

$$1_{\mathbb{Z}} := [1, 0].$$

Theorem 0.42.12 (Zero Is an Integer). *The class $0_{\mathbb{Z}}$ is an element of $\mathbb{Z}$.*

Theorem 0.42.13 (One Is an Integer). *The class $1_{\mathbb{Z}}$ is an element of $\mathbb{Z}$.*

Theorem 0.42.14 (Zero Is Not One in $\mathbb{Z}$). *In $\mathbb{Z}$,*

$$0_{\mathbb{Z}} \neq 1_{\mathbb{Z}}.$$

0.43 Operations on $\mathbb{Z}$-Classes

Addition on $\mathbb{Z}$-Classes

Definition 0.43.1 (Addition on $\mathbb{Z}$-Classes). Define addition on representatives by

$$[a, b] + [c, d] := [a + c, b + d].$$

Lemma 0.43.2 (Addition Respects Formal-Difference Equivalence on the Left). *If $[a, b] = [a', b']$, then*

$$[a, b] + [c, d] = [a', b'] + [c, d].$$

Lemma 0.43.3 (Addition Respects Formal-Difference Equivalence on the Right). *If $[c, d] = [c', d']$, then*

$$[a, b] + [c, d] = [a, b] + [c', d'].$$

Lemma 0.43.4 (Addition Respects Formal-Difference Equivalence). *If $[a, b] = [a', b']$ and $[c, d] = [c', d']$, then*

$$[a, b] + [c, d] = [a', b'] + [c', d'].$$

Theorem 0.43.5 (Addition on $\mathbb{Z}$ Is Well-Defined). *Addition determines a well-defined binary operation $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$.*

Theorem 0.43.6 (Addition on $\mathbb{Z}$ Is Associative). *For all $x, y, z \in \mathbb{Z}$,*

$$(x + y) + z = x + (y + z).$$

Theorem 0.43.7 (Addition on $\mathbb{Z}$ Is Commutative). *For all $x, y \in \mathbb{Z}$,*

$$x + y = y + x.$$

Theorem 0.43.8 (Zero Is a Left Additive Identity on $\mathbb{Z}$). *For every $x \in \mathbb{Z}$,*

$$0_{\mathbb{Z}} + x = x.$$

Corollary 0.43.9 (Zero Is a Right Additive Identity on $\mathbb{Z}$). *For every $x \in \mathbb{Z}$,*

$$x + 0_{\mathbb{Z}} = x.$$

Theorem 0.43.10 (Zero Is an Additive Identity on $\mathbb{Z}$). *For every $x \in \mathbb{Z}$,*

$$0_{\mathbb{Z}} + x = x \qquad \textit{and} \qquad x + 0_{\mathbb{Z}} = x.$$

Negation on $\mathbb{Z}$-Classes

Definition 0.43.11 (Negation on $\mathbb{Z}$-Classes)**.** Define negation on representatives by

$$-[a,b] := [b,a].$$

Lemma 0.43.12 (Negation Respects Formal-Difference Equivalence)**.** *If* $[a,b]=[a',b']$*, then*

$$-[a,b] = -[a',b'].$$

Theorem 0.43.13 (Negation on $\mathbb{Z}$ Is Well-Defined)**.** *Negation determines a well-defined unary operation* $\mathbb{Z}\to\mathbb{Z}$*.*

Theorem 0.43.14 (Negation Gives a Left Additive Inverse on $\mathbb{Z}$)**.** *For every* $x\in\mathbb{Z}$*,*

$$(-x)+x=0_{\mathbb{Z}}.$$

Corollary 0.43.15 (Negation Gives a Right Additive Inverse on $\mathbb{Z}$)**.** *For every* $x\in\mathbb{Z}$*,*

$$x+(-x)=0_{\mathbb{Z}}.$$

Theorem 0.43.16 (Every Integer Has an Additive Inverse)**.** *For every* $x\in\mathbb{Z}$ *there exists* $y\in\mathbb{Z}$ *such that*

$$x+y=0_{\mathbb{Z}} \qquad \text{and} \qquad y+x=0_{\mathbb{Z}}.$$

Subtraction on $\mathbb{Z}$-Classes

Definition 0.43.17 (Subtraction on $\mathbb{Z}$)**.** For $x,y\in\mathbb{Z}$, define

$$x-y := x+(-y).$$

Theorem 0.43.18 (Subtraction on $\mathbb{Z}$ Is Well-Defined)**.** *Subtraction determines a well-defined binary operation* $\mathbb{Z}\times\mathbb{Z}\to\mathbb{Z}$*.*

Multiplication on $\mathbb{Z}$-Classes

Definition 0.43.19 (Multiplication on $\mathbb{Z}$-Classes)**.** Define multiplication on representatives by

$$[a,b]\cdot[c,d] := [ac+bd, ad+bc].$$

Lemma 0.43.20 (Multiplication Respects Formal-Difference Equivalence on the Left)**.** *If* $[a,b]=[a',b']$*, then*

$$[a,b]\cdot[c,d]=[a',b']\cdot[c,d].$$

Lemma 0.43.21 (Multiplication Respects Formal-Difference Equivalence on the Right)**.** *If* $[c,d]=[c',d']$*, then*

$$[a,b]\cdot[c,d]=[a,b]\cdot[c',d'].$$

Lemma 0.43.22 (Multiplication Respects Formal-Difference Equivalence)**.** *If* $[a,b]=[a',b']$ *and* $[c,d]=[c',d']$*, then*

$$[a,b]\cdot[c,d]=[a',b']\cdot[c',d'].$$

Theorem 0.43.23 (Multiplication on $\mathbb{Z}$ Is Well-Defined)**.** *Multiplication determines a well-defined binary operation* $\mathbb{Z}\times\mathbb{Z}\to\mathbb{Z}$*.*

Theorem 0.43.24 (Multiplication on $\mathbb{Z}$ Is Associative)**.** *For all* $x,y,z\in\mathbb{Z}$*,*

$$(x\cdot y)\cdot z = x\cdot(y\cdot z).$$

Theorem 0.43.25 (Multiplication on $\mathbb{Z}$ Is Commutative)**.** *For all $x, y \in \mathbb{Z}$,*

$$x \cdot y = y \cdot x.$$

Theorem 0.43.26 (One Is a Left Multiplicative Identity on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$1_{\mathbb{Z}} \cdot x = x.$$

Corollary 0.43.27 (One Is a Right Multiplicative Identity on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$x \cdot 1_{\mathbb{Z}} = x.$$

Theorem 0.43.28 (One Is a Multiplicative Identity on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$1_{\mathbb{Z}} \cdot x = x \qquad \text{and} \qquad x \cdot 1_{\mathbb{Z}} = x.$$

Distributivity on $\mathbb{Z}$-Classes

Theorem 0.43.29 (Left Distributivity on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$x \cdot (y + z) = x \cdot y + x \cdot z.$$

Corollary 0.43.30 (Right Distributivity on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$(y + z) \cdot x = y \cdot x + z \cdot x.$$

Theorem 0.43.31 (Multiplication Distributes over Addition on $\mathbb{Z}$)**.** *Multiplication distributes over addition on both sides in $\mathbb{Z}$.*

Class-Level Ring Summary

Theorem 0.43.32 ($\mathbb{Z}$ Is an Additive Abelian Group)**.** *The structure $(\mathbb{Z}, +, 0_{\mathbb{Z}})$ is an abelian group.*

Theorem 0.43.33 ($\mathbb{Z}$ Is a Multiplicative Commutative Monoid)**.** *The structure $(\mathbb{Z}, \cdot, 1_{\mathbb{Z}})$ is a commutative monoid.*

Theorem 0.43.34 ($\mathbb{Z}$ Is a Commutative Ring)**.** *The structure $(\mathbb{Z}, +, \cdot, 0_{\mathbb{Z}}, 1_{\mathbb{Z}})$ is a commutative ring.*

0.44 Algebraic Structure of $\mathbb{Z}$

Integral Domain Structure

Theorem 0.44.1 ($\mathbb{Z}$ Has No Zero Divisors)**.** *For all $x, y \in \mathbb{Z}$,*

$$x \cdot y = 0 \Longrightarrow x = 0 \text{ or } y = 0.$$

Theorem 0.44.2 ($\mathbb{Z}$ Is an Integral Domain)**.** *The integers form an integral domain: $\mathbb{Z}$ is a commutative ring, $0 \neq 1$, and $\mathbb{Z}$ has no zero divisors.*

Corollary 0.44.3 (Multiplicative Cancellation on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$z \neq 0 \wedge z \cdot x = z \cdot y \Longrightarrow x = y.$$

Derived Laws on $\mathbb{Z}$

Corollary 0.44.4 (Zero Absorption on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$0 \cdot x = 0 \qquad \text{and} \qquad x \cdot 0 = 0.$$

Corollary 0.44.5 (Sign Rule on $\mathbb{Z}$)**.** *For all $x, y \in \mathbb{Z}$,*

$$(-x) \cdot y = -(x \cdot y) \qquad \text{and} \qquad x \cdot (-y) = -(x \cdot y).$$

Corollary 0.44.6 (Negative One Rule on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$(-1) \cdot x = -x.$$

Corollary 0.44.7 (Product of Negatives on $\mathbb{Z}$)**.** *For all $x, y \in \mathbb{Z}$,*

$$(-x) \cdot (-y) = x \cdot y.$$

0.45 Order on $\mathbb{Z}$

Positivity on $\mathbb{Z}$

Definition 0.45.1 (Positive Integer)**.** An integer $x \in \mathbb{Z}$ is *positive* if $x = [a, b]$ for some $a, b \in \mathbb{W}$ with $b < a$.

Lemma 0.45.2 (Positivity Respects Formal-Difference Equivalence)**.** *If $[a, b] = [a', b']$, then*

$$b < a \Longleftrightarrow b' < a'.$$

Theorem 0.45.3 (Positivity on $\mathbb{Z}$ Is Well-Defined)**.** *The predicate "x is positive" is well-defined on equivalence classes in $\mathbb{Z}$.*

Definition 0.45.4 (Strict Order on $\mathbb{Z}$)**.** For $x, y \in \mathbb{Z}$, define

$$x < y \quad :\Longleftrightarrow \quad y - x \text{ is positive.}$$

Definition 0.45.5 (Order on $\mathbb{Z}$)**.** For $x, y \in \mathbb{Z}$, define

$$x \leq y \quad :\Longleftrightarrow \quad x < y \text{ or } x = y.$$

Proposition 0.45.6 (Product of Positives Is Positive)**.** *For all $x, y \in \mathbb{Z}$,*

$$0 < x \wedge 0 < y \Longrightarrow 0 < x \cdot y.$$

Canonical Form and Sign

Theorem 0.45.7 (Canonical Form of Integers)**.** *For every $x \in \mathbb{Z}$, exactly one of the following holds:*

$$x = [n, 0] \text{ for some nonzero } n \in \mathbb{W}, \qquad x = [0, 0], \qquad x = [0, n] \text{ for some nonzero } n \in \mathbb{W}.$$

Corollary 0.45.8 (Integers Split by Sign)**.** *Every integer is nonnegative or the negative of a nonnegative integer, and the two descriptions overlap only at $0_{\mathbb{Z}}$.*

Corollary 0.45.9 (Image of the Embedding Is the Nonnegative Integers)**.** *The image of the canonical embedding $\mathbb{W} \to \mathbb{Z}$ is exactly the set of nonnegative integers.*

Theorem 0.45.10 (Sign Trichotomy on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$, exactly one of*

$$x < 0, \qquad x = 0, \qquad 0 < x$$

holds.

Theorem 0.45.11 (Trichotomy on $\mathbb{Z}$)**.** *For all $x, y \in \mathbb{Z}$, exactly one of*

$$x < y, \qquad x = y, \qquad y < x$$

holds.

Total Order on $\mathbb{Z}$

Theorem 0.45.12 (Strict Order on $\mathbb{Z}$ Is Irreflexive)**.** *For every $x \in \mathbb{Z}$, not $x < x$.*

Theorem 0.45.13 (Strict Order on $\mathbb{Z}$ Is Asymmetric)**.** *For all $x, y \in \mathbb{Z}$,*

$$x < y \implies \neg(y < x).$$

Theorem 0.45.14 (Strict Order on $\mathbb{Z}$ Is Transitive)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$x < y \wedge y < z \implies x < z.$$

Theorem 0.45.15 (Order on $\mathbb{Z}$ Is Reflexive)**.** *For every $x \in \mathbb{Z}$, $x \leq x$.*

Theorem 0.45.16 (Order on $\mathbb{Z}$ Is Antisymmetric)**.** *For all $x, y \in \mathbb{Z}$,*

$$x \leq y \wedge y \leq x \implies x = y.$$

Theorem 0.45.17 (Order on $\mathbb{Z}$ Is Transitive)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$x \leq y \wedge y \leq z \implies x \leq z.$$

Theorem 0.45.18 (Order on $\mathbb{Z}$ Is Total)**.** *For all $x, y \in \mathbb{Z}$,*

$$x \leq y \text{ or } y \leq x.$$

Theorem 0.45.19 ($\mathbb{Z}$ Is a Totally Ordered Set)**.** *The ordered pair $(\mathbb{Z}, \leq)$ is a totally ordered set.*

Addition Order Compatibility on $\mathbb{Z}$

Theorem 0.45.20 (Left Addition Preserves Strict Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$x < y \implies z + x < z + y.$$

Corollary 0.45.21 (Right Addition Preserves Strict Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$x < y \implies x + z < y + z.$$

Theorem 0.45.22 (Addition Preserves Strict Order on $\mathbb{Z}$)**.** *Addition preserves strict order on both sides in $\mathbb{Z}$.*

Theorem 0.45.23 (Left Addition Preserves Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$x \leq y \implies z + x \leq z + y.$$

Corollary 0.45.24 (Right Addition Preserves Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$x \leq y \implies x + z \leq y + z.$$

Theorem 0.45.25 (Addition Preserves Order on $\mathbb{Z}$)**.** *Addition preserves non-strict order on both sides in $\mathbb{Z}$.*

Corollary 0.45.26 (Addition Reflects Order on $\mathbb{Z}$)**.** *Addition by a fixed integer reflects both strict and non-strict order.*

Multiplication Order Compatibility on $\mathbb{Z}$

Theorem 0.45.27 (Left Multiplication by Positives Preserves Strict Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$0 < z \wedge x < y \implies z \cdot x < z \cdot y.$$

Corollary 0.45.28 (Right Multiplication by Positives Preserves Strict Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$0 < z \wedge x < y \Longrightarrow x \cdot z < y \cdot z.$$

Theorem 0.45.29 (Multiplication by Positives Preserves Strict Order on $\mathbb{Z}$)**.** *Multiplication by a positive integer preserves strict order on both sides.*

Theorem 0.45.30 (Left Multiplication by Positives Preserves Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$0 < z \wedge x \leq y \Longrightarrow z \cdot x \leq z \cdot y.$$

Corollary 0.45.31 (Right Multiplication by Positives Preserves Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$0 < z \wedge x \leq y \Longrightarrow x \cdot z \leq y \cdot z.$$

Theorem 0.45.32 (Multiplication by Positives Preserves Order on $\mathbb{Z}$)**.** *Multiplication by a positive integer preserves non-strict order on both sides.*

Theorem 0.45.33 (Multiplication by Negatives Reverses Strict Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$z < 0 \wedge x < y \Longrightarrow z \cdot y < z \cdot x.$$

Corollary 0.45.34 (Multiplication by Negatives Reverses Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$z < 0 \wedge x \leq y \Longrightarrow z \cdot y \leq z \cdot x.$$

Theorem 0.45.35 (Multiplication by Zero Collapses Order on $\mathbb{Z}$)**.** *For all $x, y \in \mathbb{Z}$,*

$$0 \cdot x = 0 \cdot y.$$

Absolute Value on $\mathbb{Z}$

Definition 0.45.36 (Absolute Value on $\mathbb{Z}$)**.** Define

$$|x| := \begin{cases} x, & 0 \leq x, \\ -x, & x < 0. \end{cases}$$

Theorem 0.45.37 (Absolute Value Is Nonnegative)**.** *For every $x \in \mathbb{Z}$,*

$$0 \leq |x|.$$

Theorem 0.45.38 (Absolute Value Vanishes Only at Zero)**.** *For every $x \in \mathbb{Z}$,*

$$|x| = 0 \Longleftrightarrow x = 0.$$

Theorem 0.45.39 (Absolute Value Is Symmetric under Negation)**.** *For every $x \in \mathbb{Z}$,*

$$|-x| = |x|.$$

Theorem 0.45.40 (Absolute Value Is Multiplicative)**.** *For all $x, y \in \mathbb{Z}$,*

$$|x \cdot y| = |x| \cdot |y|.$$

Theorem 0.45.41 (Bounding by Absolute Value)**.** *For every $x \in \mathbb{Z}$,*

$$-|x| \leq x \leq |x|.$$

Theorem 0.45.42 (Triangle Inequality on $\mathbb{Z}$)**.** *For all $x, y \in \mathbb{Z}$,*

$$|x + y| \leq |x| + |y|.$$

Well-Ordering Fails on $\mathbb{Z}$

Theorem 0.45.43 ($\mathbb{Z}$ Is Not Well-Ordered)**.** *The ordered set* $(\mathbb{Z}, \leq)$ *is not a well-order.*

0.46 Embedding $\mathbb{W}$ into $\mathbb{Z}$

Embedding $\mathbb{W}$ into $\mathbb{Z}$

Definition 0.46.1 (Embedding of $\mathbb{W}$ into $\mathbb{Z}$)**.** Define

$$\iota : \mathbb{W} \to \mathbb{Z}, \qquad \iota(n) := [n, 0].$$

Theorem 0.46.2 (Embedding Preserves Zero)**.** *The canonical embedding satisfies*

$$\iota(0_{\mathbb{W}}) = 0_{\mathbb{Z}}.$$

Theorem 0.46.3 (Embedding Preserves One)**.** *The canonical embedding satisfies*

$$\iota(1_{\mathbb{W}}) = 1_{\mathbb{Z}}.$$

Theorem 0.46.4 (Embedding $\mathbb{W}$ into $\mathbb{Z}$ Is Injective)**.** *For all* $a, b \in \mathbb{W}$*,*

$$\iota(a) = \iota(b) \Longrightarrow a = b.$$

Theorem 0.46.5 (Embedding Preserves Addition)**.** *For all* $a, b \in \mathbb{W}$*,*

$$\iota(a + b) = \iota(a) + \iota(b).$$

Theorem 0.46.6 (Embedding Preserves Multiplication)**.** *For all* $a, b \in \mathbb{W}$*,*

$$\iota(a \cdot b) = \iota(a) \cdot \iota(b).$$

Theorem 0.46.7 (Embedding Preserves Order)**.** *For all* $a, b \in \mathbb{W}$*,*

$$a \leq b \Longleftrightarrow \iota(a) \leq \iota(b).$$

Theorem 0.46.8 ($\mathbb{W}$ Embeds into $\mathbb{Z}$)**.** *The map* $\iota : \mathbb{W} \to \mathbb{Z}$ *is an injective order-preserving semiring homomorphism.*

0.47 Divisibility in $\mathbb{Z}$

Divisibility in $\mathbb{Z}$

Definition 0.47.1 (Divisibility in $\mathbb{Z}$)**.** For $a, b \in \mathbb{Z}$, define

$$a \mid b \quad :\Longleftrightarrow \quad \exists c \in \mathbb{Z}\ (b = a \cdot c).$$

Basic Divisibility Laws

Proposition 0.47.2 (Divisibility Is Reflexive on $\mathbb{Z}$)**.** *For every* $a \in \mathbb{Z}$*,*

$$a \mid a.$$

Proposition 0.47.3 (Divisibility Is Transitive on $\mathbb{Z}$)**.** *For all* $a, b, c \in \mathbb{Z}$*,*

$$a \mid b \wedge b \mid c \Longrightarrow a \mid c.$$

Proposition 0.47.4 (One and Negative One Divide Every Integer)**.** *For every $a \in \mathbb{Z}$,*

$$1 \mid a \qquad \textit{and} \qquad (-1) \mid a.$$

Proposition 0.47.5 (Every Integer Divides Zero)**.** *For every $a \in \mathbb{Z}$,*

$$a \mid 0.$$

Proposition 0.47.6 (Divisibility Is Compatible with Negation on the Divisor)**.** *For all $a, b \in \mathbb{Z}$,*

$$a \mid b \iff (-a) \mid b.$$

Proposition 0.47.7 (Divisibility Is Compatible with Negation on the Dividend)**.** *For all $a, b \in \mathbb{Z}$,*

$$a \mid b \iff a \mid (-b).$$

Proposition 0.47.8 (Divisibility Is Compatible with Negation on $\mathbb{Z}$)**.** *For all $a, b \in \mathbb{Z}$,*

$$a \mid b \iff (-a) \mid b \iff a \mid (-b).$$

Proposition 0.47.9 (Divisibility Is Compatible with Addition on $\mathbb{Z}$)**.** *For all $a, b, c \in \mathbb{Z}$,*

$$a \mid b \wedge a \mid c \Longrightarrow a \mid (b + c).$$

Proposition 0.47.10 (Divisibility Respects Linear Combinations on $\mathbb{Z}$)**.** *For all $a, b, c, x, y \in \mathbb{Z}$,*

$$a \mid b \wedge a \mid c \Longrightarrow a \mid (b \cdot x + c \cdot y).$$

Units in $\mathbb{Z}$

Definition 0.47.11 (Unit in $\mathbb{Z}$)**.** An integer $u \in \mathbb{Z}$ is a *unit* if

$$\exists v \in \mathbb{Z}\ (u \cdot v = 1).$$

Theorem 0.47.12 (The Units of $\mathbb{Z}$ Are Exactly ± 1)**.** *An integer u is a unit if and only if*

$$u = 1 \quad \textit{or} \quad u = -1.$$

Definition 0.47.13 (Associates in $\mathbb{Z}$)**.** Integers $a, b \in \mathbb{Z}$ are *associates* if $a = u \cdot b$ for some unit $u \in \mathbb{Z}$.

Division Algorithm

Theorem 0.47.14 (Division Algorithm on $\mathbb{Z}$)**.** *If $a, b \in \mathbb{Z}$ and $b \neq 0$, then there exist unique $q, r \in \mathbb{Z}$ such that*

$$a = bq + r \qquad \textit{and} \qquad 0 \leq r < |b|.$$

Greatest Common Divisor

Definition 0.47.15 (Greatest Common Divisor in $\mathbb{Z}$)**.** A greatest common divisor $\gcd(a, b)$ is the positive common divisor of a and b divisible by every other common divisor.

Bezout Identity

Theorem 0.47.16 (Bezout Identity on $\mathbb{Z}$)**.** *For $a, b \in \mathbb{Z}$, there exist $x, y \in \mathbb{Z}$ such that*

$$\gcd(a, b) = a \cdot x + b \cdot y.$$

0.48 Transition to $\mathbb{Q}$

Proofs

Capstone

Rational Numbers ($\mathbb{Q}$)

0.49 Construction of $\mathbb{Q}$

Prefractions

Definition 0.49.1 (Nonzero Integers). The set of nonzero integers is

$$\mathbb{Z}^* := \mathbb{Z} \setminus \{0_\mathbb{Z}\}.$$

Definition 0.49.2 (Prefraction). A *prefraction* is an ordered pair $(a, b) \in \mathbb{Z} \times \mathbb{Z}^*$, read as the formal quotient a/b.

Definition 0.49.3 (Prefraction Set). The prefraction set is

$$\operatorname{Pre}\mathbb{Q} := \mathbb{Z} \times \mathbb{Z}^*.$$

Fraction Equivalence

Definition 0.49.4 (Fraction Equivalence). For $(a, b), (c, d) \in \operatorname{Pre}\mathbb{Q}$, define

$$(a, b) \sim_\mathbb{Q} (c, d) \quad :\Longleftrightarrow \quad a \cdot d = b \cdot c.$$

Lemma 0.49.5 (Fraction Equivalence Is Reflexive). *For every* $(a, b) \in \operatorname{Pre}\mathbb{Q}$*,*

$$(a, b) \sim_\mathbb{Q} (a, b).$$

Lemma 0.49.6 (Fraction Equivalence Is Symmetric). *For all prefractions* $(a, b), (c, d)$*,*

$$(a, b) \sim_\mathbb{Q} (c, d) \Longrightarrow (c, d) \sim_\mathbb{Q} (a, b).$$

Lemma 0.49.7 (Fraction Equivalence Is Transitive). *For all prefractions* $(a, b), (c, d), (e, f)$*,*

$$(a, b) \sim_\mathbb{Q} (c, d) \wedge (c, d) \sim_\mathbb{Q} (e, f) \Longrightarrow (a, b) \sim_\mathbb{Q} (e, f).$$

Theorem 0.49.8 (Fraction Equivalence Is an Equivalence Relation). *The relation* $\sim_\mathbb{Q}$ *is an equivalence relation on* $\operatorname{Pre}\mathbb{Q}$*.*

Definition of $\mathbb{Q}$

Definition 0.49.9 (Rational Numbers). The rational numbers are the quotient

$$\mathbb{Q} := \operatorname{Pre}\mathbb{Q}/{\sim_\mathbb{Q}}.$$

Definition 0.49.10 (Rational Class Notation). The notation $[a, b]$ denotes the $\sim_\mathbb{Q}$-equivalence class of the prefraction (a, b), where $b \neq 0_\mathbb{Z}$.

Proposition 0.49.11 (Equality Criterion for Rational Classes). *For prefractions* $(a, b), (c, d) \in \operatorname{Pre}\mathbb{Q}$*,*

$$[a, b] = [c, d] \quad \Longleftrightarrow \quad a \cdot d = b \cdot c.$$

Zero and One in $\mathbb{Q}$

Definition 0.49.12 (Zero in $\mathbb{Q}$)**.** Define

$$0_{\mathbb{Q}} := [0_{\mathbb{Z}}, 1_{\mathbb{Z}}].$$

Definition 0.49.13 (One in $\mathbb{Q}$)**.** Define

$$1_{\mathbb{Q}} := [1_{\mathbb{Z}}, 1_{\mathbb{Z}}].$$

Lemma 0.49.14 (Vanishing Criterion in $\mathbb{Q}$)**.** *For every prefraction $(a, b) \in \operatorname{Pre}\mathbb{Q}$,*

$$[a, b] = 0_{\mathbb{Q}} \quad \iff \quad a = 0_{\mathbb{Z}}.$$

Theorem 0.49.15 (Zero Is a Rational)**.** *The class $0_{\mathbb{Q}}$ is an element of $\mathbb{Q}$.*

Theorem 0.49.16 (One Is a Rational)**.** *The class $1_{\mathbb{Q}}$ is an element of $\mathbb{Q}$.*

Theorem 0.49.17 (Zero Is Not One in $\mathbb{Q}$)**.** *In $\mathbb{Q}$,*

$$0_{\mathbb{Q}} \neq 1_{\mathbb{Q}}.$$

0.50 Operations on $\mathbb{Q}$-Classes

Addition on $\mathbb{Q}$-Classes

Definition 0.50.1 (Addition on $\mathbb{Q}$-Classes)**.** Define addition on representatives by

$$[a, b] + [c, d] := [a \cdot d + b \cdot c,\ b \cdot d].$$

Lemma 0.50.2 (Addition Lands in $\operatorname{Pre}\mathbb{Q}$)**.** *If $b \neq 0_{\mathbb{Z}}$ and $d \neq 0_{\mathbb{Z}}$, then*

$$b \cdot d \neq 0_{\mathbb{Z}}.$$

Lemma 0.50.3 (Addition Respects Fraction Equivalence on the Left)**.** *If $[a, b] = [a', b']$, then*

$$[a, b] + [c, d] = [a', b'] + [c, d].$$

Lemma 0.50.4 (Addition Respects Fraction Equivalence on the Right)**.** *If $[c, d] = [c', d']$, then*

$$[a, b] + [c, d] = [a, b] + [c', d'].$$

Lemma 0.50.5 (Addition Respects Fraction Equivalence)**.** *If $[a, b] = [a', b']$ and $[c, d] = [c', d']$, then*

$$[a, b] + [c, d] = [a', b'] + [c', d'].$$

Theorem 0.50.6 (Addition on $\mathbb{Q}$ Is Well-Defined)**.** *Addition determines a well-defined binary operation $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$.*

Theorem 0.50.7 (Addition on $\mathbb{Q}$ Is Associative)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$(x + y) + z = x + (y + z).$$

Theorem 0.50.8 (Addition on $\mathbb{Q}$ Is Commutative)**.** *For all $x, y \in \mathbb{Q}$,*

$$x + y = y + x.$$

Theorem 0.50.9 (Zero Is a Left Additive Identity on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} + x = x.$$

Corollary 0.50.10 (Zero Is a Right Additive Identity on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$x + 0_{\mathbb{Q}} = x.$$

Theorem 0.50.11 (Zero Is an Additive Identity on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} + x = x \qquad \text{and} \qquad x + 0_{\mathbb{Q}} = x.$$

Negation on $\mathbb{Q}$-Classes

Definition 0.50.12 (Negation on $\mathbb{Q}$-Classes)**.** Define negation on representatives by

$$-[a, b] := [-a, b].$$

Lemma 0.50.13 (Negation Respects Fraction Equivalence)**.** *If* $[a, b] = [a', b']$*, then*

$$-[a, b] = -[a', b'].$$

Theorem 0.50.14 (Negation on $\mathbb{Q}$ Is Well-Defined)**.** *Negation determines a well-defined unary operation* $\mathbb{Q} \to \mathbb{Q}$*.*

Theorem 0.50.15 (Negation Gives a Left Additive Inverse on $\mathbb{Q}$)**.** *For every* $x \in \mathbb{Q}$*,*

$$(-x) + x = 0_{\mathbb{Q}}.$$

Corollary 0.50.16 (Negation Gives a Right Additive Inverse on $\mathbb{Q}$)**.** *For every* $x \in \mathbb{Q}$*,*

$$x + (-x) = 0_{\mathbb{Q}}.$$

Theorem 0.50.17 (Every Rational Has an Additive Inverse)**.** *For every* $x \in \mathbb{Q}$ *there exists* $y \in \mathbb{Q}$ *such that*

$$x + y = 0_{\mathbb{Q}} \qquad \text{and} \qquad y + x = 0_{\mathbb{Q}}.$$

Subtraction on $\mathbb{Q}$

Definition 0.50.18 (Subtraction on $\mathbb{Q}$)**.** For $x, y \in \mathbb{Q}$, define

$$x - y := x + (-y).$$

Theorem 0.50.19 (Subtraction on $\mathbb{Q}$ Is Well-Defined)**.** *Subtraction determines a well-defined binary operation* $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$*.*

Multiplication on $\mathbb{Q}$-Classes

Definition 0.50.20 (Multiplication on $\mathbb{Q}$-Classes)**.** Define multiplication on representatives by

$$[a, b] \cdot [c, d] := [a \cdot c,\ b \cdot d].$$

Lemma 0.50.21 (Multiplication Lands in $\operatorname{Pre}\mathbb{Q}$)**.** *If* $b \neq 0_{\mathbb{Z}}$ *and* $d \neq 0_{\mathbb{Z}}$*, then*

$$b \cdot d \neq 0_{\mathbb{Z}}.$$

Lemma 0.50.22 (Multiplication Respects Fraction Equivalence on the Left)**.** *If* $[a, b] = [a', b']$*, then*

$$[a, b] \cdot [c, d] = [a', b'] \cdot [c, d].$$

Lemma 0.50.23 (Multiplication Respects Fraction Equivalence on the Right)**.** *If* $[c, d] = [c', d']$*, then*

$$[a, b] \cdot [c, d] = [a, b] \cdot [c', d'].$$

Lemma 0.50.24 (Multiplication Respects Fraction Equivalence)**.** *If* $[a, b] = [a', b']$ *and* $[c, d] = [c', d']$*, then*

$$[a, b] \cdot [c, d] = [a', b'] \cdot [c', d'].$$

Theorem 0.50.25 (Multiplication on $\mathbb{Q}$ Is Well-Defined)**.** *Multiplication determines a well-defined binary operation* $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$*.*

Theorem 0.50.26 (Multiplication on $\mathbb{Q}$ Is Associative). *For all $x, y, z \in \mathbb{Q}$,*

$$(x \cdot y) \cdot z = x \cdot (y \cdot z).$$

Theorem 0.50.27 (Multiplication on $\mathbb{Q}$ Is Commutative). *For all $x, y \in \mathbb{Q}$,*

$$x \cdot y = y \cdot x.$$

Theorem 0.50.28 (One Is a Left Multiplicative Identity on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$1_{\mathbb{Q}} \cdot x = x.$$

Corollary 0.50.29 (One Is a Right Multiplicative Identity on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$x \cdot 1_{\mathbb{Q}} = x.$$

Theorem 0.50.30 (One Is a Multiplicative Identity on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$1_{\mathbb{Q}} \cdot x = x \qquad \text{and} \qquad x \cdot 1_{\mathbb{Q}} = x.$$

Reciprocal on Nonzero $\mathbb{Q}$-Classes

Lemma 0.50.31 (Nonzero Predicate Is Well-Defined on $\mathbb{Q}$). *For every prefraction $(a, b) \in \operatorname{Pre}\mathbb{Q}$,*

$$[a, b] \neq 0_{\mathbb{Q}} \iff a \neq 0_{\mathbb{Z}}.$$

Thus nonzeroness is independent of the representative.

Definition 0.50.32 (Reciprocal on Nonzero $\mathbb{Q}$-Classes). For $[a, b] \neq 0_{\mathbb{Q}}$, define

$$[a, b]^{-1} := [b, a].$$

Lemma 0.50.33 (Reciprocal Lands in $\operatorname{Pre}\mathbb{Q}$). *If $[a, b] \neq 0_{\mathbb{Q}}$, then $a \neq 0_{\mathbb{Z}}$, so (b, a) is a prefraction.*

Lemma 0.50.34 (Reciprocal Respects Fraction Equivalence). *If $[a, b] = [a', b']$ and $[a, b] \neq 0_{\mathbb{Q}}$, then*

$$[a, b]^{-1} = [a', b']^{-1}.$$

Theorem 0.50.35 (Reciprocal on $\mathbb{Q}$ Is Well-Defined). *Reciprocal determines a well-defined map*

$$\mathbb{Q} \setminus \{0_{\mathbb{Q}}\} \to \mathbb{Q} \setminus \{0_{\mathbb{Q}}\}.$$

Theorem 0.50.36 (Reciprocal Gives a Left Multiplicative Inverse on $\mathbb{Q}$). *For every nonzero $x \in \mathbb{Q}$,*

$$x^{-1} \cdot x = 1_{\mathbb{Q}}.$$

Corollary 0.50.37 (Reciprocal Gives a Right Multiplicative Inverse on $\mathbb{Q}$). *For every nonzero $x \in \mathbb{Q}$,*

$$x \cdot x^{-1} = 1_{\mathbb{Q}}.$$

Theorem 0.50.38 (Every Nonzero Rational Has a Multiplicative Inverse). *For every $x \in \mathbb{Q}$ with $x \neq 0_{\mathbb{Q}}$, there exists $y \in \mathbb{Q}$ such that*

$$x \cdot y = 1_{\mathbb{Q}} \qquad \text{and} \qquad y \cdot x = 1_{\mathbb{Q}}.$$

Division on $\mathbb{Q}$

Definition 0.50.39 (Division on $\mathbb{Q}$). For $x, y \in \mathbb{Q}$ with $y \neq 0_{\mathbb{Q}}$, define

$$x/y := x \cdot y^{-1}.$$

Theorem 0.50.40 (Division on $\mathbb{Q}$ Is Well-Defined). *Division determines a well-defined map*

$$\mathbb{Q} \times (\mathbb{Q} \setminus \{0_{\mathbb{Q}}\}) \to \mathbb{Q}.$$

Distributivity on $\mathbb{Q}$-Classes

Theorem 0.50.41 (Left Distributivity on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$x \cdot (y + z) = x \cdot y + x \cdot z.$$

Corollary 0.50.42 (Right Distributivity on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$(y + z) \cdot x = y \cdot x + z \cdot x.$$

Theorem 0.50.43 (Multiplication Distributes over Addition on $\mathbb{Q}$)**.** *Multiplication distributes over addition on both sides in $\mathbb{Q}$.*

Class-Level Field Summary

Theorem 0.50.44 ($\mathbb{Q}$ Is an Additive Abelian Group)**.** *The structure $(\mathbb{Q}, +, 0_{\mathbb{Q}})$ is an abelian group.*

Theorem 0.50.45 (Nonzero $\mathbb{Q}$ Is a Multiplicative Abelian Group)**.** *The structure $(\mathbb{Q} \setminus \{0_{\mathbb{Q}}\}, \cdot, 1_{\mathbb{Q}})$ is an abelian group.*

Theorem 0.50.46 ($\mathbb{Q}$ Is a Field)**.** *The structure $(\mathbb{Q}, +, \cdot, 0_{\mathbb{Q}}, 1_{\mathbb{Q}})$ is a field.*

0.51 Field Structure of Q

Derived Field Laws on $\mathbb{Q}$

Corollary 0.51.1 (Zero Product on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$x \cdot y = 0_{\mathbb{Q}} \iff x = 0_{\mathbb{Q}} \text{ or } y = 0_{\mathbb{Q}}.$$

Corollary 0.51.2 (Double Negation on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$-(-x) = x.$$

Corollary 0.51.3 (Negation of a Product on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$(-x) \cdot y = -(x \cdot y) \qquad \text{and} \qquad x \cdot (-y) = -(x \cdot y).$$

Corollary 0.51.4 (Product of Negatives on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$(-x) \cdot (-y) = x \cdot y.$$

Corollary 0.51.5 (Additive Cancellation on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$x + z = y + z \Longrightarrow x = y.$$

Corollary 0.51.6 (Multiplicative Cancellation on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$z \neq 0_{\mathbb{Q}} \wedge x \cdot z = y \cdot z \Longrightarrow x = y.$$

Corollary 0.51.7 (Uniqueness of Additive Inverses on $\mathbb{Q}$)**.** *Every additive inverse in $\mathbb{Q}$ is unique.*

Corollary 0.51.8 (Uniqueness of Multiplicative Inverses on $\mathbb{Q}$)**.** *Every multiplicative inverse of a nonzero rational number is unique.*

Corollary 0.51.9 (The Inverse of One Is One on $\mathbb{Q}$)**.** *In $\mathbb{Q}$,*

$$1_{\mathbb{Q}}^{-1} = 1_{\mathbb{Q}}.$$

Corollary 0.51.10 (Inverse of a Product on $\mathbb{Q}$)**.** *For all nonzero $x, y \in \mathbb{Q}$,*

$$(x \cdot y)^{-1} = x^{-1} \cdot y^{-1}.$$

Corollary 0.51.11 (Inverse of an Inverse on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$,*

$$(x^{-1})^{-1} = x.$$

Integer Powers on $\mathbb{Q}$

Definition 0.51.12 (Integer Power on $\mathbb{Q}$)**.** Let $x \in \mathbb{Q}$. For $n \in \mathbb{W}$, define x^n by the usual recursive power operation. If $x \neq 0_{\mathbb{Q}}$ and $n \in \mathbb{Z}$ is negative, define

$$x^n := (x^{-1})^{-n}.$$

Corollary 0.51.13 (Exponent Addition Law on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$ and all $m, n \in \mathbb{Z}$,*

$$x^{m+n} = x^m \cdot x^n.$$

Corollary 0.51.14 (Power of a Power on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$ and all $m, n \in \mathbb{Z}$,*

$$(x^m)^n = x^{m \cdot n}.$$

Corollary 0.51.15 (Power of a Product on $\mathbb{Q}$)**.** *For all nonzero $x, y \in \mathbb{Q}$ and every $n \in \mathbb{Z}$,*

$$(x \cdot y)^n = x^n \cdot y^n.$$

0.52 Order on $\mathbb{Q}$

Positivity on $\mathbb{Q}$

Definition 0.52.1 (Positive Rational)**.** A rational number $[a, b] \in \mathbb{Q}$ is *positive* if

$$a \cdot b > 0_{\mathbb{Z}}.$$

Lemma 0.52.2 (Positivity Respects Fraction Equivalence on $\mathbb{Q}$)**.** *If $[a, b] = [c, d]$, then*

$$a \cdot b > 0_{\mathbb{Z}} \iff c \cdot d > 0_{\mathbb{Z}}.$$

Theorem 0.52.3 (Positivity on $\mathbb{Q}$ Is Well-Defined)**.** *The predicate "x is positive" is well-defined on equivalence classes in $\mathbb{Q}$.*

Definition 0.52.4 (Strict Order on $\mathbb{Q}$)**.** For $x, y \in \mathbb{Q}$, define

$$x < y \quad :\iff \quad y - x \text{ is positive.}$$

Definition 0.52.5 (Order on $\mathbb{Q}$)**.** For $x, y \in \mathbb{Q}$, define

$$x \leq y \quad :\iff \quad x < y \text{ or } x = y.$$

Proposition 0.52.6 (Product of Positives Is Positive on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x \wedge 0_{\mathbb{Q}} < y \implies 0_{\mathbb{Q}} < x \cdot y.$$

Trichotomy on $\mathbb{Q}$

Theorem 0.52.7 (Sign Trichotomy on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$, exactly one of*

$$x < 0_{\mathbb{Q}}, \qquad x = 0_{\mathbb{Q}}, \qquad 0_{\mathbb{Q}} < x$$

holds.

Theorem 0.52.8 (Trichotomy on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$, exactly one of*

$$x < y, \qquad x = y, \qquad y < x$$

holds.

Total Order on $\mathbb{Q}$

Theorem 0.52.9 (Strict Order on $\mathbb{Q}$ Is Irreflexive). *For every $x \in \mathbb{Q}$, not $x < x$.*

Theorem 0.52.10 (Strict Order on $\mathbb{Q}$ Is Asymmetric). *For all $x, y \in \mathbb{Q}$,*

$$x < y \implies \neg(y < x).$$

Theorem 0.52.11 (Strict Order on $\mathbb{Q}$ Is Transitive). *For all $x, y, z \in \mathbb{Q}$,*

$$x < y \wedge y < z \implies x < z.$$

Theorem 0.52.12 (Order on $\mathbb{Q}$ Is Reflexive). *For every $x \in \mathbb{Q}$, $x \leq x$.*

Theorem 0.52.13 (Order on $\mathbb{Q}$ Is Antisymmetric). *For all $x, y \in \mathbb{Q}$,*

$$x \leq y \wedge y \leq x \implies x = y.$$

Theorem 0.52.14 (Order on $\mathbb{Q}$ Is Transitive). *For all $x, y, z \in \mathbb{Q}$,*

$$x \leq y \wedge y \leq z \implies x \leq z.$$

Theorem 0.52.15 (Order on $\mathbb{Q}$ Is Total). *For all $x, y \in \mathbb{Q}$,*

$$x \leq y \text{ or } y \leq x.$$

Theorem 0.52.16 ($\mathbb{Q}$ Is a Totally Ordered Set). *The ordered pair $(\mathbb{Q}, \leq)$ is a totally ordered set.*

Addition Order Compatibility on $\mathbb{Q}$

Theorem 0.52.17 (Left Addition Preserves Strict Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x < y \implies z + x < z + y.$$

Corollary 0.52.18 (Right Addition Preserves Strict Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x < y \implies x + z < y + z.$$

Theorem 0.52.19 (Addition Preserves Strict Order on $\mathbb{Q}$). *Addition preserves strict order on both sides in $\mathbb{Q}$.*

Theorem 0.52.20 (Left Addition Preserves Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x \leq y \implies z + x \leq z + y.$$

Corollary 0.52.21 (Right Addition Preserves Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x \leq y \implies x + z \leq y + z.$$

Theorem 0.52.22 (Addition Preserves Order on $\mathbb{Q}$). *Addition preserves non-strict order on both sides in $\mathbb{Q}$.*

Corollary 0.52.23 (Addition Reflects Order on $\mathbb{Q}$). *Addition by a fixed rational reflects both strict and non-strict order.*

Multiplication Order Compatibility on $\mathbb{Q}$

Theorem 0.52.24 (Left Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < z \wedge x < y \implies z \cdot x < z \cdot y.$$

Corollary 0.52.25 (Right Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < z \wedge x < y \Longrightarrow x \cdot z < y \cdot z.$$

Theorem 0.52.26 (Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$)**.** *Multiplication by a positive rational preserves strict order on both sides.*

Theorem 0.52.27 (Left Multiplication by Positives Preserves Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < z \wedge x \leq y \Longrightarrow z \cdot x \leq z \cdot y.$$

Corollary 0.52.28 (Right Multiplication by Positives Preserves Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < z \wedge x \leq y \Longrightarrow x \cdot z \leq y \cdot z.$$

Theorem 0.52.29 (Multiplication by Positives Preserves Order on $\mathbb{Q}$)**.** *Multiplication by a positive rational preserves non-strict order on both sides.*

Theorem 0.52.30 (Multiplication by Negatives Reverses Strict Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$z < 0_{\mathbb{Q}} \wedge x < y \Longrightarrow z \cdot y < z \cdot x.$$

Corollary 0.52.31 (Multiplication by Negatives Reverses Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$z < 0_{\mathbb{Q}} \wedge x \leq y \Longrightarrow z \cdot y \leq z \cdot x.$$

Reciprocal Order Laws on $\mathbb{Q}$

Theorem 0.52.32 (Sign of a Reciprocal on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x \Longrightarrow 0_{\mathbb{Q}} < x^{-1}, \qquad x < 0_{\mathbb{Q}} \Longrightarrow x^{-1} < 0_{\mathbb{Q}}.$$

Theorem 0.52.33 (Reciprocal Reverses Order on Positives in $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x < y \Longrightarrow 0_{\mathbb{Q}} < y^{-1} < x^{-1}.$$

Absolute Value on $\mathbb{Q}$

Definition 0.52.34 (Absolute Value on $\mathbb{Q}$)**.** For $x \in \mathbb{Q}$, define

$$|x| := \begin{cases} x, & 0_{\mathbb{Q}} \leq x, \\ -x, & x < 0_{\mathbb{Q}}. \end{cases}$$

Theorem 0.52.35 (Absolute Value Is Nonnegative on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} \leq |x|.$$

Theorem 0.52.36 (Absolute Value Vanishes Only at Zero on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$|x| = 0_{\mathbb{Q}} \iff x = 0_{\mathbb{Q}}.$$

Theorem 0.52.37 (Absolute Value Is Multiplicative on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$|x \cdot y| = |x| \cdot |y|.$$

Theorem 0.52.38 (Triangle Inequality on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$|x + y| \leq |x| + |y|.$$

Theorem 0.52.39 (Absolute Value of a Reciprocal on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$,*

$$|x^{-1}| = |x|^{-1}.$$

0.53 Ordered Field Structure of Q

Ordered Field Instantiation

Theorem 0.53.1 (ℚ Is an Ordered Field)**.** *The structure*

$$(\mathbb{Q}, +, \cdot, 0_{\mathbb{Q}}, 1_{\mathbb{Q}}, \leq)$$

is an ordered field.

Corollary 0.53.2 (Ordered Field Laws Hold on ℚ)**.** *Every theorem that depends only on the ordered-field axioms applies to* $\mathbb{Q}$ *with its constructed operations and order.*

Standard Ordered-Field Consequences

Corollary 0.53.3 (Squares Are Nonnegative on ℚ)**.** *For every* $x \in \mathbb{Q}$*,*

$$0_{\mathbb{Q}} \leq x^2.$$

Corollary 0.53.4 (One Is Positive in ℚ)**.** *In* $\mathbb{Q}$*,*

$$0_{\mathbb{Q}} < 1_{\mathbb{Q}}.$$

Corollary 0.53.5 (Natural Numbers Are Positive in ℚ)**.** *Every natural number embedded in* $\mathbb{Q}$ *is positive.*

Corollary 0.53.6 (Positive Rationals Have Positive Inverses)**.** *For every* $x \in \mathbb{Q}$*,*

$$0_{\mathbb{Q}} < x \Longrightarrow 0_{\mathbb{Q}} < x^{-1}.$$

Corollary 0.53.7 (Fraction Comparison in ℚ)**.** *For rational classes with positive denominator product,*

$$[a, b] < [c, d] \iff a \cdot d < c \cdot b.$$

0.54 Arbitrary Closeness in ℚ

Density and the Archimedean Property

Definition 0.54.1 (Two in ℚ)**.** Define

$$2_{\mathbb{Q}} := 1_{\mathbb{Q}} + 1_{\mathbb{Q}}.$$

Lemma 0.54.2 (Two Is Nonzero in ℚ)**.** *In* $\mathbb{Q}$*,*

$$2_{\mathbb{Q}} \neq 0_{\mathbb{Q}}.$$

Lemma 0.54.3 (Midpoint Lies Strictly Between Rationals)**.** *For all* $x, y \in \mathbb{Q}$*,*

$$x < y \Longrightarrow x < (x + y) \cdot 2_{\mathbb{Q}}^{-1} < y.$$

Theorem 0.54.4 (Density of ℚ in Itself)**.** *For all* $x, y \in \mathbb{Q}$*,*

$$x < y \Longrightarrow \exists r \in \mathbb{Q}\ (x < r < y).$$

Theorem 0.54.5 (Archimedean Property of ℚ)**.** *For all* $x, y \in \mathbb{Q}$*,*

$$0_{\mathbb{Q}} < x \Longrightarrow \exists n \in \mathbb{N}\ (y < n \cdot x).$$

From Discreteness to Arbitrary Closeness

Definition 0.54.6 (Arbitrarily Close Distinct Points)**.** A number system has arbitrarily close distinct points if for every point x and every positive tolerance ε, there is a point $y \neq x$ such that

$$|x - y| < \varepsilon.$$

Theorem 0.54.7 ($\mathbb{Q}$ Has Arbitrarily Close Distinct Points)**.** *For every $x \in \mathbb{Q}$ and every $\varepsilon \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < \varepsilon \Longrightarrow \exists y \in \mathbb{Q}\ (y \neq x \text{ and } |x - y| < \varepsilon).$$

Corollary 0.54.8 ($\mathbb{Q}$ Has No Adjacent Points)**.** *There do not exist rational numbers $x < y$ with no rational number strictly between them.*

Proposition 0.54.9 ($\mathbb{Q}$ Has No Least Positive Rational)**.** *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x \Longrightarrow \exists y \in \mathbb{Q}\ (0_{\mathbb{Q}} < y < x).$$

Corollary 0.54.10 (Reciprocals Become Arbitrarily Small in $\mathbb{Q}$)**.** *For every $\varepsilon \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < \varepsilon \Longrightarrow \exists n \in \mathbb{N}\ \left(0_{\mathbb{Q}} < n^{-1} < \varepsilon\right).$$

Limit-Like Behavior Before Limits

Proposition 0.54.11 (One-Sided Rational Approximation)**.** *For every $x \in \mathbb{Q}$ and every $\varepsilon \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < \varepsilon \Longrightarrow \exists y, z \in \mathbb{Q}\ (x - \varepsilon < y < x < z < x + \varepsilon).$$

0.55 Rational Gaps

No Rational Square Root of Two

Theorem 0.55.1 (No Rational Square Root of Two)**.** *There is no rational number $q \in \mathbb{Q}$ such that*

$$q \cdot q = 2_{\mathbb{Q}}.$$

The Square-Two Gap

Definition 0.55.2 (The Gap Set in $\mathbb{Q}$)**.** Define

$$S := \{q \in \mathbb{Q} : 0_{\mathbb{Q}} < q \text{ and } q \cdot q < 2_{\mathbb{Q}}\}.$$

Proposition 0.55.3 (The Gap Set Is Bounded Above in $\mathbb{Q}$)**.** *The set S is nonempty and bounded above in $\mathbb{Q}$.*

Theorem 0.55.4 (The Square-Two Gap Has No Rational Supremum)**.** *The set S has no least upper bound in $\mathbb{Q}$.*

Corollary 0.55.5 ($\mathbb{Q}$ Is Order-Incomplete)**.** *The ordered field $\mathbb{Q}$ is not order-complete: there exists a nonempty bounded-above subset of $\mathbb{Q}$ with no supremum in $\mathbb{Q}$.*

Cauchy Sequences in $\mathbb{Q}$

Definition 0.55.6 (Cauchy Sequence in $\mathbb{Q}$)**.** A sequence (x_n) in $\mathbb{Q}$ is Cauchy if

$$\forall \varepsilon \in \mathbb{Q}\ (0_{\mathbb{Q}} < \varepsilon \Longrightarrow \exists N \in \mathbb{N}\, \forall m, n \geq N\ |x_m - x_n| < \varepsilon).$$

Definition 0.55.7 (Null Sequence in $\mathbb{Q}$)**.** A sequence (x_n) in $\mathbb{Q}$ is null if

$$\forall \varepsilon \in \mathbb{Q}\ (0_{\mathbb{Q}} < \varepsilon \Longrightarrow \exists N \in \mathbb{N}\, \forall n \geq N\ |x_n| < \varepsilon).$$

Lemma 0.55.8 (Rational Limits Are Unique)**.** *A convergent sequence in $\mathbb{Q}$ has at most one rational limit.*

0.56 Embedding $\mathbb{Z}$ into $\mathbb{Q}$

The Integer Embedding

Definition 0.56.1 (Embedding of $\mathbb{Z}$ into $\mathbb{Q}$)**.** Define

$$\iota : \mathbb{Z} \to \mathbb{Q}, \qquad \iota(n) := [n, 1_{\mathbb{Z}}].$$

Theorem 0.56.2 (Embedding Preserves Zero)**.** *The embedding sends integer zero to rational zero:*

$$\iota(0_{\mathbb{Z}}) = 0_{\mathbb{Q}}.$$

Theorem 0.56.3 (Embedding Preserves One)**.** *The embedding sends integer one to rational one:*

$$\iota(1_{\mathbb{Z}}) = 1_{\mathbb{Q}}.$$

Theorem 0.56.4 (Embedding $\mathbb{Z}$ into $\mathbb{Q}$ Is Injective)**.** *For all $m, n \in \mathbb{Z}$,*

$$\iota(m) = \iota(n) \implies m = n.$$

Preservation of Algebra and Order

Theorem 0.56.5 (Embedding Preserves Addition)**.** *For all $m, n \in \mathbb{Z}$,*

$$\iota(m + n) = \iota(m) + \iota(n).$$

Theorem 0.56.6 (Embedding Preserves Multiplication)**.** *For all $m, n \in \mathbb{Z}$,*

$$\iota(m \cdot n) = \iota(m) \cdot \iota(n).$$

Theorem 0.56.7 (Embedding Preserves Order)**.** *For all $m, n \in \mathbb{Z}$,*

$$m \leq n \iff \iota(m) \leq \iota(n).$$

Theorem 0.56.8 ($\mathbb{Z}$ Embeds into $\mathbb{Q}$)**.** *The map $\iota : \mathbb{Z} \to \mathbb{Q}$ is an injective order-preserving ring homomorphism.*

Field of Fractions

Theorem 0.56.9 ($\mathbb{Q}$ Is the Field of Fractions of $\mathbb{Z}$)**.** *For every $x \in \mathbb{Q}$, there exist $m, n \in \mathbb{Z}$ with $n \neq 0_{\mathbb{Z}}$ such that*

$$x = \iota(m) \cdot \iota(n)^{-1}.$$

Corollary 0.56.10 (The Embedding of $\mathbb{Z}$ into $\mathbb{Q}$ Is Not Surjective)**.** *The image of $\iota : \mathbb{Z} \to \mathbb{Q}$ is a proper subset of $\mathbb{Q}$.*

0.57 Transition to $\mathbb{R}$

Why the Real Numbers Are Introduced

0.58 Canonical Rational Number Statement Plan

Proofs

Real Numbers ($\mathbb{R}$)

0.59 Construction of $\mathbb{R}$

Dedekind Cuts

Definition 0.59.1 (Dedekind Cut)**.** A *Dedekind cut* is a set $\alpha \subseteq \mathbb{Q}$ satisfying:

$$\alpha \neq \emptyset, \qquad \alpha \neq \mathbb{Q},$$

$$p \in \alpha \wedge q < p \Longrightarrow q \in \alpha,$$

and

$$p \in \alpha \Longrightarrow \exists r \in \alpha\,(p < r).$$

Definition 0.59.2 (The Real Numbers)**.** The real numbers are the set

$$\mathbb{R} := \{\alpha : \alpha \text{ is a Dedekind cut in } \mathbb{Q}\}.$$

Rational Cuts

Definition 0.59.3 (Rational Cut)**.** For $q \in \mathbb{Q}$, define the rational cut

$$q^* := \{r \in \mathbb{Q} : r < q\}.$$

Lemma 0.59.4 (Rational Cuts Are Cuts)**.** *For every $q \in \mathbb{Q}$, the set q^* is a Dedekind cut.*

Zero and One in $\mathbb{R}$

Definition 0.59.5 (Zero in $\mathbb{R}$)**.** Define

$$0_{\mathbb{R}} := (0_{\mathbb{Q}})^* = \{r \in \mathbb{Q} : r < 0_{\mathbb{Q}}\}.$$

Definition 0.59.6 (One in $\mathbb{R}$)**.** Define

$$1_{\mathbb{R}} := (1_{\mathbb{Q}})^* = \{r \in \mathbb{Q} : r < 1_{\mathbb{Q}}\}.$$

Theorem 0.59.7 (Zero and One Are Reals)**.** *The cuts $0_{\mathbb{R}}$ and $1_{\mathbb{R}}$ are elements of $\mathbb{R}$.*

Theorem 0.59.8 (Zero Is Not One in $\mathbb{R}$)**.** *In $\mathbb{R}$,*

$$0_{\mathbb{R}} \neq 1_{\mathbb{R}}.$$

0.60 Order on $\mathbb{R}$

Order by Inclusion

Definition 0.60.1 (Order on $\mathbb{R}$)**.** For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha \leq \beta \quad :\Longleftrightarrow \quad \alpha \subseteq \beta.$$

Definition 0.60.2 (Strict Order on $\mathbb{R}$)**.** For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha < \beta \quad :\Longleftrightarrow \quad \alpha \subsetneq \beta.$$

Theorem 0.60.3 (Order on $\mathbb{R}$ Is Reflexive)**.** *For every $\alpha \in \mathbb{R}$, $\alpha \leq \alpha$.*

Theorem 0.60.4 (Order on $\mathbb{R}$ Is Antisymmetric)**.** *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha \leq \beta \wedge \beta \leq \alpha \Longrightarrow \alpha = \beta.$$

Theorem 0.60.5 (Order on $\mathbb{R}$ Is Transitive)**.** *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$\alpha \leq \beta \wedge \beta \leq \gamma \Longrightarrow \alpha \leq \gamma.$$

Theorem 0.60.6 (Comparability of Cuts)**.** *For all cuts $\alpha, \beta \in \mathbb{R}$,*

$$\alpha \subseteq \beta \vee \beta \subseteq \alpha.$$

Theorem 0.60.7 (Order on $\mathbb{R}$ Is Total)**.** *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha \leq \beta \vee \beta \leq \alpha.$$

Theorem 0.60.8 (Trichotomy on $\mathbb{R}$)**.** *For all $\alpha, \beta \in \mathbb{R}$, exactly one of*

$$\alpha < \beta, \qquad \alpha = \beta, \qquad \beta < \alpha$$

holds.

Theorem 0.60.9 ($\mathbb{R}$ Is a Totally Ordered Set)**.** *The ordered pair $(\mathbb{R}, \leq)$ is a total order.*

0.61 Completeness of $\mathbb{R}$

Definition 0.61.1 (Upper Bound in $\mathbb{R}$)**.** Let $S \subseteq \mathbb{R}$. A cut $\gamma \in \mathbb{R}$ is an upper bound of S if

$$\forall \alpha \in S \, (\alpha \leq \gamma).$$

The set S is bounded above if it has an upper bound in $\mathbb{R}$.

Definition 0.61.2 (Supremum in $\mathbb{R}$)**.** For $S \subseteq \mathbb{R}$, a cut σ is the supremum of S if σ is an upper bound of S and every upper bound γ of S satisfies $\sigma \leq \gamma$.

Definition 0.61.3 (Least-Upper-Bound Property on $\mathbb{R}$)**.** The ordered set $\mathbb{R}$ has the *least-upper-bound property* if every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

Lemma 0.61.4 (Union of a Bounded Nonempty Family of Cuts Is a Cut)**.** *If $S \subseteq \mathbb{R}$ is nonempty and bounded above, then*

$$\bigcup_{\alpha \in S} \alpha$$

is a Dedekind cut.

Theorem 0.61.5 (Least-Upper-Bound Property)**.** *Every nonempty subset $S \subseteq \mathbb{R}$ that is bounded above has a least upper bound, namely*

$$\sup S = \bigcup_{\alpha \in S} \alpha.$$

Corollary 0.61.6 (Greatest-Lower-Bound Property)**.** *Every nonempty subset $S \subseteq \mathbb{R}$ that is bounded below has a greatest lower bound.*

0.62 Addition on $\mathbb{R}$

Definition 0.62.1 (Addition of Cuts)**.** For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha + \beta := \{p + q : p \in \alpha,\ q \in \beta\}.$$

Lemma 0.62.2 (Sum of Cuts Is a Cut)**.** *If $\alpha, \beta \in \mathbb{R}$, then $\alpha + \beta$ is a Dedekind cut.*

Theorem 0.62.3 (Addition on $\mathbb{R}$ Is Associative)**.** *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma).$$

Theorem 0.62.4 (Addition on $\mathbb{R}$ Is Commutative)**.** *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha + \beta = \beta + \alpha.$$

Theorem 0.62.5 (Zero Is an Additive Identity on $\mathbb{R}$)**.** *For every $\alpha \in \mathbb{R}$,*

$$\alpha + 0_{\mathbb{R}} = \alpha.$$

Definition 0.62.6 (Negation of a Cut)**.** For $\alpha \in \mathbb{R}$, define

$$-\alpha := \{p \in \mathbb{Q} : \exists r \in \mathbb{Q}\,(r > 0 \wedge -p - r \notin \alpha)\}.$$

Lemma 0.62.7 (Negation of a Cut Is a Cut)**.** *If $\alpha \in \mathbb{R}$, then $-\alpha$ is a Dedekind cut.*

Theorem 0.62.8 (Negation Is an Additive Inverse on $\mathbb{R}$)**.** *For every $\alpha \in \mathbb{R}$,*

$$\alpha + (-\alpha) = 0_{\mathbb{R}}.$$

Theorem 0.62.9 ($\mathbb{R}$ Is an Additive Abelian Group)**.** *The structure $(\mathbb{R}, +, 0_{\mathbb{R}})$ is an abelian group.*

Definition 0.62.10 (Subtraction on $\mathbb{R}$)**.** For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha - \beta := \alpha + (-\beta).$$

0.63 Multiplication on $\mathbb{R}$

Definition 0.63.1 (Nonnegative Cut)**.** A cut $\alpha \in \mathbb{R}$ is nonnegative if $0_{\mathbb{R}} \leq \alpha$ and positive if $0_{\mathbb{R}} < \alpha$.

Definition 0.63.2 (Product of Nonnegative Cuts)**.** For nonnegative cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha \cdot \beta := \{r \in \mathbb{Q} : r < 0_{\mathbb{Q}}\} \cup \{p \cdot q : p \in \alpha,\ q \in \beta,\ p \geq 0_{\mathbb{Q}},\ q \geq 0_{\mathbb{Q}}\}.$$

Lemma 0.63.3 (Product of Nonnegative Cuts Is a Cut)**.** *If $\alpha, \beta \geq 0_{\mathbb{R}}$, then $\alpha \cdot \beta$ is a Dedekind cut.*

Definition 0.63.4 (Absolute Value on $\mathbb{R}$)**.** For $\alpha \in \mathbb{R}$, define

$$|\alpha| := \begin{cases} \alpha, & 0_{\mathbb{R}} \leq \alpha, \\ -\alpha, & \alpha < 0_{\mathbb{R}}. \end{cases}$$

Theorem 0.63.5 (Triangle Inequality)**.** *For all $\alpha, \beta \in \mathbb{R}$,*

$$|\alpha + \beta| \leq |\alpha| + |\beta|.$$

Go to proof.

Definition 0.63.6 (Multiplication of Cuts)**.** For cuts $\alpha, \beta \in \mathbb{R}$, define $\alpha \cdot \beta$ by sign cases: use $|\alpha| \cdot |\beta|$ when α and β have the same sign, use $-(|\alpha| \cdot |\beta|)$ when they have opposite signs, and use $0_{\mathbb{R}}$ when either factor is $0_{\mathbb{R}}$.

Lemma 0.63.7 (Product of Cuts Is a Cut). *If $\alpha, \beta \in \mathbb{R}$, then $\alpha \cdot \beta \in \mathbb{R}$.*

Theorem 0.63.8 (Multiplication on $\mathbb{R}$ Is Associative). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$(\alpha \cdot \beta) \cdot \gamma = \alpha \cdot (\beta \cdot \gamma).$$

Theorem 0.63.9 (Multiplication on $\mathbb{R}$ Is Commutative). *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha \cdot \beta = \beta \cdot \alpha.$$

Theorem 0.63.10 (One Is a Multiplicative Identity on $\mathbb{R}$). *For every $\alpha \in \mathbb{R}$,*

$$\alpha \cdot 1_{\mathbb{R}} = \alpha.$$

Definition 0.63.11 (Reciprocal of a Positive Cut). For $\alpha > 0_{\mathbb{R}}$, define

$$\alpha^{-1} := \{r \in \mathbb{Q} : r \leq 0_{\mathbb{Q}}\} \cup \{r \in \mathbb{Q} : r > 0_{\mathbb{Q}} \land \exists s \in \mathbb{Q}\, (s > r \land s^{-1} \notin \alpha)\}.$$

Definition 0.63.12 (Reciprocal of a Nonzero Cut). For $\alpha \neq 0_{\mathbb{R}}$, define α^{-1} to be $|\alpha|^{-1}$ if $\alpha > 0_{\mathbb{R}}$ and $-(|\alpha|^{-1})$ if $\alpha < 0_{\mathbb{R}}$.

Lemma 0.63.13 (Reciprocal of a Nonzero Cut Is a Cut). *If $\alpha \neq 0_{\mathbb{R}}$, then $\alpha^{-1} \in \mathbb{R}$.*

Theorem 0.63.14 (Reciprocal Is a Multiplicative Inverse on $\mathbb{R}$). *If $\alpha \neq 0_{\mathbb{R}}$, then*

$$\alpha \cdot \alpha^{-1} = 1_{\mathbb{R}}.$$

Definition 0.63.15 (Division on $\mathbb{R}$). For $\beta \neq 0_{\mathbb{R}}$, define

$$\alpha/\beta := \alpha \cdot \beta^{-1}.$$

Theorem 0.63.16 (Distributivity on $\mathbb{R}$). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma.$$

0.64 Field and Ordered-Field Structure of $\mathbb{R}$

Theorem 0.64.1 (Nonzero $\mathbb{R}$ Is a Multiplicative Abelian Group). *The structure $(\mathbb{R} \setminus \{0_{\mathbb{R}}\}, \cdot, 1_{\mathbb{R}})$ is an abelian group.*

Theorem 0.64.2 ($\mathbb{R}$ Is a Field). *The structure $(\mathbb{R}, +, \cdot, 0_{\mathbb{R}}, 1_{\mathbb{R}})$ is a field.*

Corollary 0.64.3 (Field Laws Hold on $\mathbb{R}$). *Every theorem proved generically for fields holds in $\mathbb{R}$, including zero product, double negation, sign rules, cancellation, uniqueness of inverses, inverse laws, and exponent laws.*

Order Compatibility

Theorem 0.64.4 (Addition Preserves Order on $\mathbb{R}$). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$\alpha < \beta \Longrightarrow \alpha + \gamma < \beta + \gamma,$$

and the corresponding non-strict statement also holds.

Theorem 0.64.5 (Multiplication by Positives Preserves Order on $\mathbb{R}$). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$0_{\mathbb{R}} < \gamma \land \alpha < \beta \Longrightarrow \gamma \cdot \alpha < \gamma \cdot \beta.$$

Theorem 0.64.6 ($\mathbb{R}$ Is an Ordered Field). *The structure $(\mathbb{R}, +, \cdot, 0_{\mathbb{R}}, 1_{\mathbb{R}}, \leq)$ is an ordered field.*

Corollary 0.64.7 (Ordered-Field Laws Hold on $\mathbb{R}$). *Every theorem proved generically for ordered fields holds in $\mathbb{R}$.*

Theorem 0.64.8 ($\mathbb{R}$ Is a Complete Ordered Field). *The ordered field $\mathbb{R}$ satisfies the least-upper-bound property.*

0.65 Embedding $\mathbb{Q}$ into $\mathbb{R}$

Definition 0.65.1 (Embedding of $\mathbb{Q}$ into $\mathbb{R}$)**.** Define $\iota : \mathbb{Q} \to \mathbb{R}$ by

$$\iota(q) := q^*.$$

Theorem 0.65.2 (Embedding Preserves Zero and One)**.** *The embedding satisfies*

$$\iota(0_\mathbb{Q}) = 0_\mathbb{R} \qquad \textit{and} \qquad \iota(1_\mathbb{Q}) = 1_\mathbb{R}.$$

Theorem 0.65.3 (Embedding $\mathbb{Q}$ into $\mathbb{R}$ Is Injective)**.** *For all $p, q \in \mathbb{Q}$,*

$$\iota(p) = \iota(q) \Longrightarrow p = q.$$

Theorem 0.65.4 (Embedding Preserves Addition and Multiplication)**.** *For all $p, q \in \mathbb{Q}$,*

$$\iota(p+q) = \iota(p) + \iota(q) \qquad \textit{and} \qquad \iota(p \cdot q) = \iota(p) \cdot \iota(q).$$

Theorem 0.65.5 (Embedding Preserves Order)**.** *For all $p, q \in \mathbb{Q}$,*

$$p \le q \Longleftrightarrow \iota(p) \le \iota(q).$$

Theorem 0.65.6 ($\mathbb{Q}$ Embeds into $\mathbb{R}$ as an Ordered Field)**.** *The map ι is an injective order-preserving field homomorphism.*

Theorem 0.65.7 (Density of $\mathbb{Q}$ in $\mathbb{R}$)**.** *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha < \beta \Longrightarrow \exists q \in \mathbb{Q}\, (\alpha < \iota(q) < \beta).$$

Theorem 0.65.8 (Archimedean Property of $\mathbb{R}$)**.** *For $\alpha, \beta \in \mathbb{R}$,*

$$0_\mathbb{R} < \alpha \Longrightarrow \exists n \in \mathbb{N}\, (\beta < \iota(n) \cdot \alpha).$$

0.66 The Gap Filled

Definition 0.66.1 (The Square-Root-of-Two Cut)**.** Define

$$\sqrt{2} := \{r \in \mathbb{Q} : r < 0_\mathbb{Q} \vee r \cdot r < 2_\mathbb{Q}\}.$$

Lemma 0.66.2 (The Square-Root-of-Two Cut Is a Cut)**.** *The set $\sqrt{2}$ is a Dedekind cut.*

Theorem 0.66.3 (Its Square Is Two)**.** *In $\mathbb{R}$,*

$$\sqrt{2} \cdot \sqrt{2} = 2_\mathbb{R},$$

where $2_\mathbb{R} := \iota(2_\mathbb{Q})$.

Corollary 0.66.4 ($\mathbb{R}$ Contains the Supremum $\mathbb{Q}$ Was Missing)**.** *Let*

$$S = \{q \in \mathbb{Q} : 0_\mathbb{Q} < q \wedge q \cdot q < 2_\mathbb{Q}\}.$$

Then

$$\sqrt{2} = \sup \iota(S).$$

Thus the bounded-above rational set with no rational supremum has a supremum in $\mathbb{R}$.

0.67 Characterization of $\mathbb{R}$

Definition 0.67.1 (Complete Ordered Field)**.** A *complete ordered field* is an ordered field satisfying the least-upper-bound property.

Theorem 0.67.2 (Uniqueness of the Complete Ordered Field)**.** *Any two complete ordered fields are isomorphic by a unique order-isomorphism.*

Corollary 0.67.3 ($\mathbb{R}$ Is the Real Numbers)**.** *The field $\mathbb{R}$ is determined up to unique order-isomorphism by being a complete ordered field.*

0.68 Transition and Forward

Proofs

Capstone

Complex Numbers

0.69 Construction of $\mathbb{C}$

Complex Prepairs

Definition 0.69.1 (Complex Prepair). A *complex prepair* is an ordered pair $(a, b) \in \mathbb{R} \times \mathbb{R}$, read informally as the expression $a + bi$.

Definition 0.69.2 (Complex Prepair Set). The complex prepair set is

$$\operatorname{Pre}\mathbb{C} := \mathbb{R} \times \mathbb{R}.$$

Coordinate Equivalence

Definition 0.69.3 (Complex Coordinate Equivalence). For $(a, b), (c, d) \in \operatorname{Pre}\mathbb{C}$, define

$$(a, b) \sim_{\mathbb{C}} (c, d) \quad :\Longleftrightarrow \quad a = c \text{ and } b = d.$$

Lemma 0.69.4 (Complex Coordinate Equivalence Is Reflexive). *For every* $(a, b) \in \operatorname{Pre}\mathbb{C}$*,*

$$(a, b) \sim_{\mathbb{C}} (a, b).$$

Lemma 0.69.5 (Complex Coordinate Equivalence Is Symmetric). *For all prepairs* $(a, b), (c, d)$*,*

$$(a, b) \sim_{\mathbb{C}} (c, d) \Longrightarrow (c, d) \sim_{\mathbb{C}} (a, b).$$

Lemma 0.69.6 (Complex Coordinate Equivalence Is Transitive). *For all prepairs* $(a, b), (c, d), (e, f)$*,*

$$(a, b) \sim_{\mathbb{C}} (c, d) \wedge (c, d) \sim_{\mathbb{C}} (e, f) \Longrightarrow (a, b) \sim_{\mathbb{C}} (e, f).$$

Theorem 0.69.7 (Complex Coordinate Equivalence Is an Equivalence Relation). *The relation* $\sim_{\mathbb{C}}$ *is an equivalence relation on* $\operatorname{Pre}\mathbb{C}$*.*

Definition of $\mathbb{C}$

Definition 0.69.8 (Complex Numbers). The complex numbers are the quotient

$$\mathbb{C} := \operatorname{Pre}\mathbb{C}/{\sim_{\mathbb{C}}}.$$

Definition 0.69.9 (Complex Class Notation). The notation $[a, b]_{\mathbb{C}}$ denotes the $\sim_{\mathbb{C}}$-equivalence class of the prepair (a, b).

Definition 0.69.10 (Zero, One, and the Imaginary Unit in $\mathbb{C}$). Define

$$0_{\mathbb{C}} := [0, 0]_{\mathbb{C}}, \qquad 1_{\mathbb{C}} := [1, 0]_{\mathbb{C}}, \qquad i := [0, 1]_{\mathbb{C}}.$$

0.70 Operations on $\mathbb{C}$-Classes

Addition on $\mathbb{C}$-Classes

Definition 0.70.1 (Addition on $\mathbb{C}$-Classes)**.** Define addition on representatives by

$$[a, b]_{\mathbb{C}} + [c, d]_{\mathbb{C}} := [a + c, b + d]_{\mathbb{C}}.$$

Lemma 0.70.2 (Addition Respects Complex Coordinate Equivalence)**.** *If $[a, b]_{\mathbb{C}} = [a', b']_{\mathbb{C}}$ and $[c, d]_{\mathbb{C}} = [c', d']_{\mathbb{C}}$, then*

$$[a, b]_{\mathbb{C}} + [c, d]_{\mathbb{C}} = [a', b']_{\mathbb{C}} + [c', d']_{\mathbb{C}}.$$

Theorem 0.70.3 (Addition on $\mathbb{C}$ Is Well-Defined)**.** *Addition determines a well-defined binary operation*

$$\mathbb{C} \times \mathbb{C} \to \mathbb{C}.$$

Negation and Subtraction

Definition 0.70.4 (Negation on $\mathbb{C}$-Classes)**.** Define negation by

$$-[a, b]_{\mathbb{C}} := [-a, -b]_{\mathbb{C}}.$$

Lemma 0.70.5 (Negation Respects Complex Coordinate Equivalence)**.** *If $[a, b]_{\mathbb{C}} = [a', b']_{\mathbb{C}}$, then*

$$-[a, b]_{\mathbb{C}} = -[a', b']_{\mathbb{C}}.$$

Theorem 0.70.6 (Negation on $\mathbb{C}$ Is Well-Defined)**.** *Negation determines a well-defined unary operation $\mathbb{C} \to \mathbb{C}$.*

Definition 0.70.7 (Subtraction on $\mathbb{C}$)**.** For $z, w \in \mathbb{C}$, define

$$z - w := z + (-w).$$

Theorem 0.70.8 (Subtraction on $\mathbb{C}$ Is Well-Defined)**.** *Subtraction determines a well-defined binary operation $\mathbb{C} \times \mathbb{C} \to \mathbb{C}$.*

Multiplication on $\mathbb{C}$-Classes

Definition 0.70.9 (Multiplication on $\mathbb{C}$-Classes)**.** Define multiplication on representatives by

$$[a, b]_{\mathbb{C}} \cdot [c, d]_{\mathbb{C}} := [ac - bd, ad + bc]_{\mathbb{C}}.$$

Lemma 0.70.10 (Multiplication Respects Complex Coordinate Equivalence)**.** *If $[a, b]_{\mathbb{C}} = [a', b']_{\mathbb{C}}$ and $[c, d]_{\mathbb{C}} = [c', d']_{\mathbb{C}}$, then*

$$[a, b]_{\mathbb{C}} \cdot [c, d]_{\mathbb{C}} = [a', b']_{\mathbb{C}} \cdot [c', d']_{\mathbb{C}}.$$

Theorem 0.70.11 (Multiplication on $\mathbb{C}$ Is Well-Defined)**.** *Multiplication determines a well-defined binary operation*

$$\mathbb{C} \times \mathbb{C} \to \mathbb{C}.$$

Conjugation and Inverses

Definition 0.70.12 (Complex Conjugation)**.** For $z = [a, b]_{\mathbb{C}}$, define its *complex conjugate* by

$$\overline{z} := [a, -b]_{\mathbb{C}}.$$

Theorem 0.70.13 (Complex Conjugation Is Well-Defined)**.** *Complex conjugation determines a well-defined unary operation $\mathbb{C} \to \mathbb{C}$.*

Lemma 0.70.14 (Product with the Conjugate Is Real and Nonnegative). *For $z = [a, b]_{\mathbb{C}}$,*

$$z\overline{z} = [a^2 + b^2, 0]_{\mathbb{C}}.$$

Moreover, if $z \neq 0_{\mathbb{C}}$, then $a^2 + b^2 \neq 0$.

Definition 0.70.15 (Multiplicative Inverse in $\mathbb{C}$). For $z = [a, b]_{\mathbb{C}} \neq 0_{\mathbb{C}}$, define

$$z^{-1} := \left[\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2}\right]_{\mathbb{C}}.$$

Theorem 0.70.16 (Inversion on $\mathbb{C}^\times$ Is Well-Defined). *Inversion determines a well-defined operation on the nonzero complex numbers.*

0.71 Field Structure of $\mathbb{C}$

Additive Structure

Theorem 0.71.1 (Addition on $\mathbb{C}$ Is Associative). *For all $z, w, u \in \mathbb{C}$,*

$$(z + w) + u = z + (w + u).$$

Theorem 0.71.2 (Addition on $\mathbb{C}$ Is Commutative). *For all $z, w \in \mathbb{C}$,*

$$z + w = w + z.$$

Theorem 0.71.3 (Zero Is an Additive Identity on $\mathbb{C}$). *For every $z \in \mathbb{C}$,*

$$0_{\mathbb{C}} + z = z \qquad \text{and} \qquad z + 0_{\mathbb{C}} = z.$$

Theorem 0.71.4 (Every Complex Number Has an Additive Inverse). *For every $z \in \mathbb{C}$,*

$$z + (-z) = 0_{\mathbb{C}} \qquad \text{and} \qquad (-z) + z = 0_{\mathbb{C}}.$$

Multiplicative Structure

Theorem 0.71.5 (Multiplication on $\mathbb{C}$ Is Associative). *For all $z, w, u \in \mathbb{C}$,*

$$(zw)u = z(wu).$$

Theorem 0.71.6 (Multiplication on $\mathbb{C}$ Is Commutative). *For all $z, w \in \mathbb{C}$,*

$$zw = wz.$$

Theorem 0.71.7 (One Is a Multiplicative Identity on $\mathbb{C}$). *For every $z \in \mathbb{C}$,*

$$1_{\mathbb{C}}z = z \qquad \text{and} \qquad z1_{\mathbb{C}} = z.$$

Theorem 0.71.8 (Every Nonzero Complex Number Has a Multiplicative Inverse). *For every $z \in \mathbb{C}$ with $z \neq 0_{\mathbb{C}}$,*

$$zz^{-1} = 1_{\mathbb{C}} \qquad \text{and} \qquad z^{-1}z = 1_{\mathbb{C}}.$$

Distributivity and the Field Theorem

Theorem 0.71.9 (Multiplication Distributes over Addition on $\mathbb{C}$). *For all $z, w, u \in \mathbb{C}$,*

$$z(w + u) = zw + zu \qquad \text{and} \qquad (w + u)z = wz + uz.$$

Theorem 0.71.10 ($\mathbb{C}$ Is a Field). *With the operations defined above,*

$$(\mathbb{C}, +, \cdot, 0_{\mathbb{C}}, 1_{\mathbb{C}})$$

is a field.

The Imaginary Unit

Theorem 0.71.11 (The Imaginary Unit Squares to Negative One)**.** *In* $\mathbb{C}$*,*

$$i^2 = -1_{\mathbb{C}}.$$

Theorem 0.71.12 ($\mathbb{C}$ Is Not an Ordered Field)**.** *There is no order relation on* $\mathbb{C}$ *making* $\mathbb{C}$ *an ordered field and extending the usual order on* $\mathbb{R}$*.*

0.72 Embedding $\mathbb{R}$ into $\mathbb{C}$

The Real Axis

Definition 0.72.1 (Canonical Embedding of $\mathbb{R}$ into $\mathbb{C}$)**.** Define

$$\iota : \mathbb{R} \to \mathbb{C}, \qquad \iota(a) := [a, 0]_{\mathbb{C}}.$$

Theorem 0.72.2 (The Real Embedding into $\mathbb{C}$ Is Injective)**.** *If* $\iota(a) = \iota(b)$*, then* $a = b$*.*

Theorem 0.72.3 (The Real Embedding Preserves Addition and Multiplication)**.** *For all* $a, b \in \mathbb{R}$*,*

$$\iota(a + b) = \iota(a) + \iota(b), \qquad \iota(ab) = \iota(a)\iota(b).$$

Theorem 0.72.4 (The Real Embedding Preserves Zero and One)**.**

$$\iota(0) = 0_{\mathbb{C}}, \qquad \iota(1) = 1_{\mathbb{C}}.$$

Real and Imaginary Parts

Definition 0.72.5 (Real and Imaginary Parts)**.** For $z = [a, b]_{\mathbb{C}}$, define

$$\mathrm{Re}(z) := a, \qquad \mathrm{Im}(z) := b.$$

Theorem 0.72.6 (Real and Imaginary Parts Are Well-Defined)**.** *The functions*

$$\mathrm{Re}, \mathrm{Im} : \mathbb{C} \to \mathbb{R}$$

are well-defined.

Proofs

Capstone

Number Lines and Intervals

0.73 Number Lines

Definition 0.73.1 (Number Line)**.** A *number line* for a linearly ordered system $(A, <_A)$ is a linearly ordered set of points L together with an order isomorphism $\lambda : A \to L$, so that for all $a, a' \in A$,

$$a <_A a' \iff \lambda(a) \text{ lies to the left of } \lambda(a').$$

0.74 Intervals on a Number Line

Definition 0.74.1 (Open Interval)**.** Let $a, b \in \mathbb{R}$ with $a \leq b$. The *open interval* with endpoints a and b is

$$(a, b) = \{\, x \in \mathbb{R} : a < x < b \,\}.$$

Definition 0.74.2 (Closed Interval)**.** Let $a, b \in \mathbb{R}$ with $a \leq b$. The *closed interval* with endpoints a and b is

$$[a, b] = \{\, x \in \mathbb{R} : a \leq x \leq b \,\}.$$

Definition 0.74.3 (Left-Half-Open Interval)**.** Let $a, b \in \mathbb{R}$ with $a \leq b$. The *left-half-open interval* with endpoints a and b is

$$(a, b] = \{\, x \in \mathbb{R} : a < x \leq b \,\}.$$

Definition 0.74.4 (Right-Half-Open Interval)**.** Let $a, b \in \mathbb{R}$ with $a \leq b$. The *right-half-open interval* with endpoints a and b is

$$[a, b) = \{\, x \in \mathbb{R} : a \leq x < b \,\}.$$

Definition 0.74.5 (Ray)**.** Let $a \in \mathbb{R}$. The four *rays* determined by a are

$$(a, \infty) = \{x : x > a\},\ [a, \infty) = \{x : x \geq a\},\ (-\infty, a) = \{x : x < a\},\ (-\infty, a] = \{x : x \leq a\}.$$

Proofs

Capstone

Embedding the Number Systems

0.75 Embeddings as Structure Preserving Maps

Definition 0.75.1 (Structure-Preserving Map)**.** Let A and B be ordered arithmetic systems with a specified common source structure: selected constants, selected operations, and a strict order. A map $\varphi : A \to B$ is *structure-preserving* exactly when it preserves each selected constant and, for every selected binary operation $\star$ and all $a, a' \in A$,

$$\varphi(a \star_A a') = \varphi(a) \star_B \varphi(a'),$$

and $a <_A a' \implies \varphi(a) <_B \varphi(a')$.

Definition 0.75.2 (Embedding of Number Systems)**.** Let A and B be ordered arithmetic systems. A map $\varphi : A \to B$ is an *embedding* exactly when φ is injective and structure-preserving and is an order embedding.

Definition 0.75.3 (Image of an Embedding)**.** Let $\varphi : A \to B$ be an embedding. The *image* of φ is the set $\varphi(A) = \{\, \varphi(a) : a \in A \,\} \subseteq B$, equipped with the selected constants, operations, and order of B restricted to $\varphi(A)$.

Theorem 0.75.4 (Embeddings Identify a System with Its Image)**.** *Let $\varphi : A \to B$ be an embedding of ordered arithmetic systems relative to a selected source structure. Then the corestriction $\varphi : A \to \varphi(A)$ is an isomorphism onto its image: it is a bijection that preserves and reflects the selected constants, operations, and order. Consequently A may be identified with the corresponding substructure $\varphi(A) \subseteq B$.* *Go to proof.*

0.76 Embedding N Into W

Definition 0.76.1 (Embedding $\mathbb{N}$ into $\mathbb{W}$)**.** The *canonical embedding* $\iota : \mathbb{N} \to \mathbb{W}$ is defined by

$$\iota(n) = n \quad \text{(the copy of each natural number inside the whole numbers).}$$

Theorem 0.76.2 ($\mathbb{N}$ Embeds in $\mathbb{W}$)**.** *The canonical map $\iota : \mathbb{N} \to \mathbb{W}$ identifies $\mathbb{N}$ with the positive part $\mathbb{W} \setminus \{0\}$ and preserves successor,* 1*, addition, multiplication, and order.* *Go to proof.*

0.77 Embedding W Into Z

Definition 0.77.1 (Embedding $\mathbb{W}$ into $\mathbb{Z}$)**.** The *canonical embedding* $\iota : \mathbb{W} \to \mathbb{Z}$ is defined by

$$\iota(w) = [(w, 0)] \quad \text{(the integer represented by } w \text{ and } 0).$$

Theorem 0.77.2 ($\mathbb{W}$ Embeds in $\mathbb{Z}$)**.** *The canonical map $\iota : \mathbb{W} \to \mathbb{Z}$ is an embedding of ordered arithmetic systems: it is injective, preserves* 0, 1*, addition, and multiplication, and reflects the order.* *Go to proof.*

0.78 Embedding Z Into Q

Definition 0.78.1 (Embedding $\mathbb{Z}$ into $\mathbb{Q}$)**.** The *canonical embedding* $\iota : \mathbb{Z} \to \mathbb{Q}$ is defined by

$$\iota(z) = [(z, 1)] \quad \text{(the rational represented by } z/1).$$

Theorem 0.78.2 ($\mathbb{Z}$ Embeds in $\mathbb{Q}$)**.** *The canonical map $\iota : \mathbb{Z} \to \mathbb{Q}$ is an embedding of ordered arithmetic systems: it is injective, preserves* 0, 1, *addition, and multiplication, and reflects the order.* *Go to proof.*

0.79 Embedding Q Into R

Definition 0.79.1 (Embedding $\mathbb{Q}$ into $\mathbb{R}$)**.** The *canonical embedding* $\iota : \mathbb{Q} \to \mathbb{R}$ is defined by

$$\iota(q) = q^* \quad \text{(the real number determined by } q\text{)}.$$

Theorem 0.79.2 ($\mathbb{Q}$ Embeds in $\mathbb{R}$)**.** *The canonical map $\iota : \mathbb{Q} \to \mathbb{R}$ is an embedding of ordered arithmetic systems: it is injective, preserves* 0, 1, *addition, and multiplication, and reflects the order.* *Go to proof.*

0.80 Compatibility of Arithmetic and Order

Theorem 0.80.1 (Compatibility of the Embedding Chain)**.** *Let $\iota_1 : \mathbb{N} \to \mathbb{W}$, $\iota_2 : \mathbb{W} \to \mathbb{Z}$, $\iota_3 : \mathbb{Z} \to \mathbb{Q}$, and $\iota_4 : \mathbb{Q} \to \mathbb{R}$ be the canonical embeddings. For every $n \in \mathbb{N}$ the composite $(\iota_4 \circ \iota_3 \circ \iota_2 \circ \iota_1)(n)$ is the real number identified with n, and the composite preserves the arithmetic and order carried by the source system. Hence arithmetic computed in any system agrees with arithmetic on its image in $\mathbb{R}$.* *Go to proof.*

0.81 One Chain Many Structures

Proofs

Capstone

Summary: The Number Systems

0.82 Axiom Systems for $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$

0.83 Number Systems Summary

The Natural Numbers $\mathbb{N}$

The Whole Numbers $\mathbb{W}$

The Integers $\mathbb{Z}$

Derived Properties of $\mathbb{Z}$ (from the Ring Axioms)

Structural Comparison: $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$

Proofs

Capstone

Origins of Numbers

Lean 4 Formalization

0.84 A First Lean 4 Reading